



## Flexi Mock CAT - 09 (2020)

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VARC

LRDI

QA

## Sec 1

**Direction for questions (1-4):** Read the given passage and answer the questions that follow.

Having more folds in this part of the brain is important because those folds increase the surface area available for brain cells, or neurons, without making the brain too big for the skull. Demonstrating that the human gene fulfills a similar purpose in the brain of another primate provides new insight into how humans may have evolved and may point the way to future treatments for brain disease.

"The human-specific sequence of new B is absolutely essential for the ability of the gene to amplify the relevant brain stem cells in development," says Wieland Huttner of the Max Planck Institute of Molecular Cell Biology, one of the study's authors.

Previous studies showed similar effects in mice and ferrets modified to have the new *B* gene. However, using those animal models meant the gene was not necessarily expressed in the same way it is in humans. Study author Michael Heide, also of the Max Planck Institute, says the team wanted to study a model organism closely related to humans, and the two most practical options were the marmoset and the macaque.

"We thought that the marmoset would be the better model because the macaque neocortex has many features that it shares with our big and folded neocortex. However, the marmoset is smooth and very small in size." Thus, any changes to the size and shape of the marmoset neocortex would be easy to see.

To introduce the gene into monkey embryos, the researchers used a "lentivirus," a virus carrier that cannot replicate. The lentivirus contained *ARHGAP11B* as well as a protein marker that would allow the researchers to see where that gene was expressed. They included a promoter gene, or a DNA sequence that regulates expression of specific genes.

Debra Silver, an investigator at the Duke University Institute for Brain Sciences, says the researchers' methods in this study, improved from those used with mice and ferrets, lend a lot of weight to the significance of the results. "One of the challenges [for this kind of study] is you can have abnormally high levels [of expression]. It's like taking a Mack truck to drive something versus something subtler like a Toyota. The idea is, with this they're trying to get closer to what would be normally expressed in the human brain."

"We have a whole list of genes that we think might be important in what makes us uniquely human, but very rarely have we definitively shown that they actually are contributors," Dennis says. "And I have to say that a study like this really brings *ARHGAP11B* up to the top of the list as a gene that could very well be important in human brain development."

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**Q.1 [11594329]**

Why did the researchers prefer a marmoset over a macaque for the study?

- 
- 1 ☐ Marmoset is a smooth and tiny creature as opposed to macaque.
- 
- 2 ☐ Marmoset neocortex more closely resembles the folded human neocortex than macaque neocortex does.
- 
- 3 ☐ It is easier to observe changes to the size and shape of a marmoset neocortex as compared to a macaque neocortex.
- 
- 4 ☐ The gene expresses itself better in the neocortex of a marmoset than in the neocortex of a macaque.
-



**Solution:**

**Correct Answer : 3**

**Your Answer : 3**

 Answer key/Solution

**Refer Para 5. The reason behind rejecting macaque was that it closely resembled the big and folded human neocortex. So 2 can't be the answer. 1 is not the immediate reason. Because of marmoset's being smooth and small, it would be easier to observe changes to its neocortex. And that is why it was chosen. Hence 3, and not 1. 4 is not mentioned in the passage.**

Bookmark

FeedBack

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## Q.2 [11594329]

Which of the following information, if true, would most strengthen the results of the study?

- 1 ☐ After the new B gene was introduced in the marmosets, it was found to control the production of certain neurons that are responsible for higher-order processing.
  - 2 ☐ The new B gene, once introduced in the marmosets, was passed on successfully to the subsequent generations.
  - 3 ☐ Once the new B gene was introduced, the folds in the neocortices of marmosets kept on growing as their age increased.
-

4 ○ A subsequent study found another gene in the neocortex of humans, similar to new B in its structure, that helped them to plan for the future.

**Solution:**

**Correct Answer : 1**

 Answer key/Solution

**1 strengthens the result of the study because it shows that not only does new B increase the folds of the neocortex, it also controls those neurons in neocortex that make it the centre of higher cognitive functions. 2 just shows the evolutionary prowess of the gene, and not its importance for human brain development. 3 talks about the continuous growth of folds. It does not directly comment on new B's role in higher cognitive functions. Hence, 1 is better. 4 shows that some other gene similar to new B could be responsible for neocortex being the centre of higher cognitive functions. In a way, it weakens the results.**

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### Q.3 [11594329]

Why does Debra Silver say “It’s like taking a Mack truck to drive something versus something subtler like a Toyota.”?

- 1 ☐ To highlight the difference in the intensity and thus the experience of driving the two vehicles.
  - 2 ☐ Because for Debra Silver, Mack Truck symbolizes abnormally high levels of gene expression whereas Toyota symbolizes normal levels of gene expression.
  - 3 ☐ To assert through an analogy that by choosing marmosets, the study was able to get a level of gene expression that was closer to that in humans.
-

4 ○ To state that the research methodology of the study under discussion lends a lot of weight to the significance of its results.



**Solution:**

**Correct Answer : 3**

**Your Answer : 4**

**The purpose of Debra Silver is not to discuss vehicles, so 1 can be eliminated.**

**While 2 is not an entirely incorrect statement, it just elaborates on what they stand for, for Debra, and not why she made their reference in the present context. 3 describes the purpose accurately. 4 is not the immediate purpose behind Debra's statement. She makes her statement in reference to the methodology only, but specifically in the context of gene expression.**

 Answer key/Solution

Bookmark

FeedBack

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#### **Q.4 [11594329]**

Which of the following are the critics of the study most likely to cite to attack its results?

- 1 ☐ In very rare cases, it has been found that the lentivirus used by researchers can replicate and mutate.
  - 2 ☐ In the studies on mice and ferrets, the new B gene was expressed in exactly the same way as that in humans.
  - 3 ☐ The promoter gene used by the researchers was significantly responsible for the increase in the folds in the brain of marmosets.
-



4 ☐ Because a macaque's brain is very closely related to the human brain, new B expresses itself in macaque's brain in an almost the same manner as that in humans.

**Solution:**

**Correct Answer : 3**

 Answer key/Solution

If the promoter gene was significantly responsible for the increase in the folds in the brain, that means new B was responsible only in a minor way, thus showing that it has at best only a minor contribution to human brain development. Hence 3. 1 is rejected because of the rarity of the event. 2 does not weaken the results of the study under discussion. 4 just states that the study could have used a closer animal model. It doesn't weaken the conclusions that the current study has arrived at.

Bookmark

FeedBack

**Direction for questions (5-9):** Read the given passage and answer the questions that follow.

Editors at *Merriam-Webster* are working on a revision of the definition of *racism*. So, the Great Awakening is even going so far as to change the dictionary? Not quite—sociopolitics drew the usage of the word *racism* beyond the dictionary definition long ago, and it is high time our dictionaries got the message.

LIKE OTHERS, THE *Merriam-Webster* dictionary has, up to now, given us what we might consider the 1.0 definition of *racism*, the one we would cite for the curious child. That is, what used to be referred to as *prejudiced*: “a belief that race is the primary determinant of human traits and capacities and that racial differences produce an inherent superiority of a particular race.”

Since the 1960s, however, *racism* has often been used in terms such as *societal racism* and *institutional racism*, referring to structures of society that disadvantage people of subordinated races because of the collective effect of bigoted attitudes. So one might say that societal racism is to blame for neighborhoods with decaying infrastructure, because white flight lowered tax revenues.

These terms have naturally often been shortened to just *racism*, such that the word has acquired a 2.0 definition. *Merriam-Webster* captures this as well, noting that *racism* can mean “a political or social system founded on racism.”

However, Kennedy Mitchum, 22 and just out of college, wrote a message to the editors at *Merriam-Webster* asking them to expand the definition to account for usage that has morphed even beyond the 2.0 definition to refer to “social and institutional power.” Mitchum noted that racism “is a system of advantage based on skin color.”

Here, the focus of the definition is less on attitudes than results: The societal disparities between white people and others are themselves referred to as racism, as a kind of shorthand for the attitudinal racism *creating* the disparities. MITCHUM WAS FRUSTRATED by people telling her, in debates about racism, that her 3.0 definition was erroneous given that it wasn’t “what’s in the dictionary.” Her frustration was justified. She wasn’t creating her own definition of the word—it is shared by legions of people, especially educated ones, across our nation.

Dictionaries can lag behind societal developments, and the idea that a “word” indisputably “means” what dictionaries say is simply sloppy. Words’ meanings change inevitably and constantly, and not just in terms of slang. As such, it won’t do for definitions of words as crucial as *racism* to keep sitting somewhere around the era of Watergate and fondue.

To extend a word’s definition from a personal quality to a society is, in itself, hardly unusual. The progression of *racism* 1.0 to *racism* 2.0 follows a line of metaphorical reasoning common as far back as the Greek philosophers treating societies as individuals in macrocosm. The conceptual step between a healthy person and a healthy society is a short one, as is that between a racist person and a racist society.

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### Q.5 [11594329]

Which of the following is the author most likely to agree with?

- 1 ☐ Merriam Webster's revision of its definition of racism symbolizes the 'Great Awakening' going far into the realm of linguistics.
- 2 ☐ There is a difference between societal and institutional racism and social and institutional power.
- 3 ☐ Dictionaries tend to lag behind societal developments.
- 4 ☐ The definition of racism proposed by Kennedy Mitchum is already used by everyone.



**Solution:**

**Correct Answer : 2**

**Your Answer : 3**

**Refer Para 1. The author denies that the current revision is not the 'Great Awakening' going very far, but the dictionary catching up with the current usage. The author will agree with 2 as societal and institutional racism is the focus of the 2.0 definition of racism, while social and institutional power is the focus of 3.0. 3 is eliminated because of 'tend to'. The author says dictionaries 'can', not 'tend to'. 4 is eliminated because while that definition is shared by "legions of people", we can't expand that to include everyone.**

 Answer key/Solution

Bookmark

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#### Q.6 [11594329]

According to the passage which of the following is true?

- 1 ☐ Greek philosophers treated the society as the whole of a structure of which individuals were a part.
- 2 ☐ Kennedy Mitchum was frustrated because people were accusing her of being unoriginal in her definition.
- 3 ☐ It is incontestable that racial disparities arise from bias-infused barriers.
- 4 ☐ Racism is a system of advantage based on skin color.



**Solution:**

**Correct Answer : 1**

**Your Answer : 1**

**4 is the 3.0 definition proposed by Kennedy and the editors of Merriam Webster are working on revising their definition of racism based on it. Also, this definition is shared by a vast number of people. This is still not enough to call it true. It is in the nature of an opinion, and the author calls the change to this definition as "less typical" in the last para. 2 is factually false. The author says that option 3 is implied by the 3.0 definition. He doesn't call it true. 1 is true as it is stated in the second last para.**

[🔍 Answer key/Solution](#)

Bookmark

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### Q.7 [11594329]

Based on the passage, what does the term “the era of Watergate and fondue” imply?

- 1 ☐ A period when racial discrimination was as mainstream as Watergate and fondue.
- 2 ☐ A period of the past when the 2.0 definition of racism came into being.
- 3 ☐ A period when the 3.0 definition of racism was not present in dictionaries.
- 4 ☐ A period when people believed that the meaning of a word is simply what the dictionaries say.

**Solution:****Correct Answer : 3****Your Answer : 4**[🔍 Answer key/Solution](#)

**All that can be gathered from Para 7 is that this era is a period of the past (“keep sitting”). And as the dictionaries are currently incorporating the 3.0 definition, which is a progression from 2.0, it means it wasn’t there in that era. Hence, 3. 1 is out of scope. 2 can’t be inferred because the para does not state the exact years of “Watergate and fondue”. 4 is distorted as the author hasn’t related this belief to any particular period.**

[Bookmark](#)[FeedBack](#)

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**Q.8 [11594329]**

Which of the following best captures the essence of the sentence "The conceptual step between a healthy person and a healthy society is a short one, as is that between a racist person and a racist society."?

- 1 ☐ A society and its individuals are closely entwined in terms of their health conditions and racist attitudes, so it is reasonable to infer about one by looking at the other.
- 2 ☐ As individuals make up a society, it is reasonable to conclude that the conditions and attitudes of the influential among them extend to become those of the whole society.
- 3 ☐ As a society is a collection of individuals, it is reasonable to extend the health conditions or racist attitudes of the society to its individuals.
- 4 ☐ As individuals are a representative part of a society, it is reasonable to extend the condition or attitudes of its individuals to the society as a whole.

**Solution:**

**Correct Answer : 4**

**Refer the second last para. The first sentence clearly talks about extending a personal quality to a society. So, 1 and 3 can be eliminated. 2 is eliminated because it talks about 'influential' individuals, which is out of scope.**

 Answer key/Solution

Bookmark

FeedBack

**Direction for questions (5-9):** Read the given passage and answer the questions that follow.

Editors at *Merriam-Webster* are working on a revision of the definition of *racism*. So, the Great Awakening is even going so far as to change the dictionary? Not quite—sociopolitics drew the usage of the word *racism* beyond the dictionary definition long ago, and it is high time our dictionaries got the message.

LIKE OTHERS, THE *Merriam-Webster* dictionary has, up to now, given us what we might consider the 1.0 definition of *racism*, the one we would cite for the curious child. That is, what used to be referred to as *prejudiced*: “a belief that race is the primary determinant of human traits and capacities and that racial differences produce an inherent superiority of a particular race.”

Since the 1960s, however, *racism* has often been used in terms such as *societal racism* and *institutional racism*, referring to structures of society that disadvantage people of subordinated races because of the collective effect of bigoted attitudes. So one might say that societal racism is to blame for neighborhoods with decaying infrastructure, because white flight lowered tax revenues.

These terms have naturally often been shortened to just *racism*, such that the word has acquired a 2.0 definition. *Merriam-Webster* captures this as well, noting that *racism* can mean “a political or social system founded on racism.”

However, Kennedy Mitchum, 22 and just out of college, wrote a message to the editors at *Merriam-Webster* asking them to expand the definition to account for usage that has morphed even beyond the 2.0 definition to refer to “social and institutional power.” Mitchum noted that racism “is a system of advantage based on skin color.”

Here, the focus of the definition is less on attitudes than results: The societal disparities between white people and others are themselves referred to as racism, as a kind of shorthand for the attitudinal racism *creating* the disparities. MITCHUM WAS FRUSTRATED by people telling her, in debates about racism, that her 3.0 definition was erroneous given that it wasn’t “what’s in the dictionary.” Her frustration was justified. She wasn’t creating her own definition of the word—it is shared by legions of people, especially educated ones, across our nation.

Dictionaries can lag behind societal developments, and the idea that a “word” indisputably “means” what dictionaries say is simply sloppy. Words’ meanings change inevitably and constantly, and not just in terms of slang. As such, it won’t do for definitions of words as crucial as *racism* to keep sitting somewhere around the era of Watergate and fondue.

To extend a word’s definition from a personal quality to a society is, in itself, hardly unusual. The progression of *racism* 1.0 to *racism* 2.0 follows a line of metaphorical reasoning common as far back as the Greek philosophers treating societies as individuals in macrocosm. The conceptual step between a healthy person and a healthy society is a short one, as is that between a racist person and a racist society.

But the step is only short when the idea is still that discrimination, standing at the gate, is the essence of racism. The progression from *racism* 2.0 to *racism* 3.0 is less typical as language change goes. The 3.0 usage implies that calling racial disparities “racism” is natural because it is indisputable that racial disparities stem from bias-infused barriers.

**Q.9 [11594329]**

What is the role played by the sentence "So one might say that societal racism is to blame for neighborhoods with decaying infrastructure, because white flight lowered tax revenues."?

- 1 ☐ It is an opinion with which the author does not agree as it places the blame for the present condition of a neighbourhood on white people who no longer live there.
- 2 ☐ It serves to explain societal racism as a practice by which one section of the society deemed superior can create structures that collectively disadvantage another section.
- 3 ☐ It illustrates how collective prejudice towards the races considered inferior results in societal structures that handicap them.
- 4 ☐ It serves to exemplify the ghettoization of subordinated races as a result of societal structures which exhibit collective bigoted attitudes towards them.

**Solution:**

**Correct Answer : 3**

**Refer Para 3. The given sentence is an example of societal racism, as defined in the passage. So 1 and 2 can be eliminated. 1, because there is no contradiction in the passage to show that the author disagrees with the statement. 2, because the statement is an example, rather than an elaboration. Also, prejudice is missing from 2. Between 3 and 4, 3 is the clear answer as the statement talks about decaying infrastructure which is a disadvantage/handicap. 4 is eliminated because there is no discussion of ghettoization or restricting the minorities to a particular area.**

 Answer key/Solution

Bookmark

FeedBack

**Direction for questions (10-14):** Read the given passage and answer the questions that follow.

What will economic recovery in the face of Coronavirus look like? At this fraught moment, no one knows enough about consumer sentiment and government ordinances and business failures and stimulus packages and the spread of the disease to make solid predictions about the future. The Trump administration and some bullish financial forecasters are arguing that we will end up with a strong, V-shaped rebound, with economic activity surging right back to where it was in no time. Others are betting on a longer, slower, U-shaped turnaround, with the pain extending for a year or three. Still others are sketching out a kind of flaccid check mark, its long tail sagging torpid into the future.

Faced with the historic catastrophe, the United States marshalled a historic response: Republicans in the White House and Congress, generally hostile to the notion of economic stimulus for low-income households, came together with Democrats to achieve a \$2 trillion rescue package, including a \$1,200 onetime payment for most adults and \$500 for many children, a radical expansion of the unemployment-insurance system to include gig workers, and a \$600-a-week bump to unemployment-insurance payouts. It also created a sweeping small-business rescue plan, covering payroll for companies that kept their employees on the books.

The good: This money kept families afloat—at least for the first, intense months of shelter-in-place. New estimates suggest that the Congressional rescue plan prevented poverty rates from rising, with many jobless workers seeing their incomes increase during lockdown due to the expanded unemployment-insurance payouts. The bad: It left out roughly 15 million people in immigrant families, many of whom were working essential jobs stocking grocery shelves, delivering takeout, and drawing blood in hospitals. And the ugly: The big helicopter drop was a onetime thing, and the unemployment-insurance expansion was time-limited. Congress designed Uncle Sam's help to dry up this summer, with the unemployment rate still in the double digits. Democrats and Republicans are negotiating another stimulus bill, but concerns about surging budget deficits are complicating the talks.

That means households are headed for a cliff. But not everyone will be affected by it equally. Rich workers, the ones with do-anywhere office jobs, have remained relatively untouched by job and earnings losses thus far. Wealthy families have seen their stock portfolios rebound to close to where they were in the winter. But poor workers—disproportionately black and Latino workers, as well as younger workers—have borne the heaviest employment and earnings losses. They entered this recession with no wealth cushion, many saddled with heavy rents and heavy debts. Income and job losses for them translate into a loss of demand economy-wide, absent federal intervention.

If and when that federal intervention dries up, millions of families just keeping their head above water will sink, as lost jobs and cancelled hours force them to stop paying their rent and go into arrears on their debt payments. Hunger, homelessness, forgotten plans to attend community college, babies growing up in stressed households: These are the stakes. The CBO forecasts that every quarter through the end of 2021, American consumers will buy \$300 billion to \$370 billion less than they would have if the pandemic had never happened. [...]

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**Q.10 [11594329]**

Based on Para 1, which of the following appears to be the most miserable prediction about the future of the economy?

---

1 ☐ A strong V-shaped rebound

---

2 ☐ A flaccid check mark

---

3 ☐ A U-shaped turnaround

4 ☐ Both 2 and 3 are equally miserable.



**Solution:**

**Correct Answer : 2**

**Your Answer : 2**

 Answer key/Solution

The para clearly states, “a kind of flaccid check mark, its long tail sagging torpid into the future.” The words sagging implies hanging down, torpid implies inactive, and most importantly ‘long’ implies that the scenario will continue for a long period of time. 1 can be easily eliminated as it is a ‘strong’ rebound. 3 says that the pain will extend “for a year or three’, thus indicating that it would not be a very long-term scenario.

Bookmark

FeedBack

**Direction for questions (10-14):** Read the given passage and answer the questions that follow.

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**Q.11 [11594329]**

What does the word 'cliff' as used in Para 4 imply?

- 
- 1 ☐ A financial crossroads that will require the households to make tough decisions.
- 
- 2 ☐ A financial fall that will not affect all households equally, with the rich and do-anywhere job holders likely to be cushioned.
-

3 ○ A financially hazardous situation affecting households unevenly.

4 ○ A financial roadblock that would not affect all households in the same way, with the rich and do-anywhere job holders probably navigating it.



**Solution:**

**Correct Answer : 3**

**Your Answer : 3**

**A cliff here signifies a hazardous situation, a sort of edge or brink, but not necessarily a fall. So 2 is rejected. 1 is eliminated as the households are not facing any two or more tough choices. 4 is rejected because the analogy between a cliff and a roadblock is not correct. A person wishes to move beyond a roadblock but wishes to recede from a cliff. 3 is the most appropriate contextual meaning.**

 Answer key/Solution

Bookmark

FeedBack

**Direction for questions (10-14):** Read the given passage and answer the questions that follow.

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**Q.12 [11594329]**

Which of the following, if true, would most weaken the argument mainly presented by the author in the last paragraph?

- 1 ☐ The author, instead of appealing to logic, tries to appeal to the emotion of the reader by painting a miserable picture of hunger, homelessness, stressed households, etc.



2 ☐ The CBO is known to take the most bearish scenario while making its forecasts and the real decrease in spending may be less.

3 ☐ The federal stimulus being negotiated is focused on measures for considerable job creation and continual monetary payments to the affected households till such jobs are created.

4 ☐ The poor households should re-skill or up-skill themselves, so that they can find alternative jobs to pay their rent and debt.



**Solution:**

**Correct Answer : 3**

**Your Answer : 3**

 Answer key/Solution

**1 is in the nature of a judgement. The author has cited facts in support of her argument. Hunger, homelessness, etc. are real possibilities in the absence of a savings cushion and rent and debt commitments. 2 weakens the CBO's forecast, but it is not weakening the argument in a major way. 3 does that job by stating that the stimulus would continue till the jobs are created, thus ensuring that the situation of "sinking" does not arise. 4 is in the nature of an advice and does not affect the real outcomes.**

Bookmark

FeedBack

**Direction for questions (10-14):** Read the given passage and answer the questions that follow.

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**Q.13 [11594329]**

From the passage, all of the following can be inferred EXCEPT:

- 
- 1 ☐ The term 'Uncle Sam' refers to the US federal government.
- 
- 2 ☐ Even though the unknown variables in the current pandemic situation make it difficult to make concrete predictions about the future, some predictions are being made.
-

3 ☐ Immigrant families in the US primarily work in jobs such as grocery stockers, takeout delivery personnel, and frontline hospital staff.

4 ☐ The term 'helicopter drop' has been used metaphorically in the passage to refer to the monetary stimulus.



**Solution:**

**Correct Answer : 3**

**Your Answer : 3**

**1 and 4 can be easily inferred from Para 3. 2 can be inferred from the entire first paragraph and also the last sentence of the passage. 3 cannot be inferred as Para 3 simply states "many of whom were working essential jobs" mentioned in Option 3. We cannot, however, generalize this statement to mean that these are the primary sectors that the immigrants "primarily" work in. It is both broad and extreme.**

 Answer key/Solution

Bookmark

FeedBack

**Direction for questions (10-14):** Read the given passage and answer the questions that follow.

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**Q.14 [11594329]**

Based on the author's views in the passage, she is likely to agree with all of the following EXCEPT?

- 1 ☐ The government's stimulus should be more inclusive to cover even the immigrants.
  - 2 ☐ Some people in the US have seen their incomes rise for the short term during the pandemic.
-

3 ☐ People of some races and ethnicities in the US have been affected more than the others during the pandemic.

4 ☐ The government should expand the unemployment-insurance indefinitely.



**Solution:**

**Correct Answer : 4**

**Your Answer : 4**

**4 is an extreme position because of 'indefinitely' and the author does not advocate any such extreme position in the passage. 1 can be inferred from Para 3. 2 is a fact stated by the author in Para 3. 3 is again a fact stated by the author in Para 4.**

 Answer key/Solution

Bookmark

FeedBack

**Direction for questions (15-18):** Read the given passage and answer the questions that follow.

Winston Churchill famously called Indians “a beastly people with a beastly religion who bred like rabbits”.

Madhushree Mukerjee’s Churchill’s Secret War establishes how Churchill and his associates could easily have stopped the Bengal famine of 1943 with a few shipments of food grains but refused, in spite of repeated appeals from two successive Viceroys, Churchill’s own Secretary of State for India and even the President of the United States.

Famines, never unknown in India, became increasingly lethal during the Raj because of the export of food grains and the replacement of food crops with indigo or jute. The Second World War made things worse, especially after Japanese forces occupied Burma in 1942, cutting off Indian rice imports. Then a destructive cyclone hit the Bengal coast just when the crucial winter crop was maturing, and the surviving rice was damaged by disease. Officials of the Raj, fearing a Japanese invasion, confiscated everything that might help the invading force – boats, carts, motor vehicles, elephants and, crucially, all the rice available. The Japanese never came but a panicking public – and many crafty businessmen – immediately began to hoard rice and the staple food of the people quickly disappeared from the marketplace.

Government stocks were released but only to feed the people of Calcutta and the urban population never suffered too greatly. The rural masses, however, were left to the wolves. This was when Churchill could have made a difference by sending wheat or rice to Bengal, and not enormous quantities. The point was to make hoarding unprofitable and as the Viceroy Lord Linlithgow pointed out, “the mere knowledge of impending imports” would have done so by lowering the price of rice.

Churchill and his war cabinet, however, decided to reserve available shipping to take food to Italy in case it fell to the Allies. He even refused help from Subhas Chandra Bose, then fighting with Axis forces, and also from Australia and Canada.

So, hundreds of thousands perished in the villages of Bengal. In London, Churchill’s beloved advisor, the physicist Frederick Alexander Lindemann (Lord Cherwell), was unmoved. A firm believer in Malthusian population theory, he blamed Indian philoprogenitiveness for the famine – sending more food would worsen the situation by encouraging Indians to breed more. The prime minister was of the same opinion and expressed himself so colourfully that Leo Amery, Secretary of State for India, exploded at him, comparing his attitudes to Hitler’s.

The Churchill industry has always denied that their idol could have done anything to relieve the Bengal famine. Shipping, they claim, was scarce and it just wasn’t possible to send food to Bengal. Mukerjee nails those “terminological inexactitudes” with precision. There was a shipping glut in summer and autumn 1943, thanks to the US transferring cargo ships to British control. Churchill, Lindemann and their close associates simply did not consider Indian lives worth saving. [...]

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**Q.15 [11594329]**

According to the passage, what led to the hoarding of rice in India?

- 
- 1 ☐ The fear of a Japanese invasion as Japanese had already invaded Burma in 1942
- 
- 2 ☐ The destructive cyclone that hit the Bengal coast and the resulting damage of rice by disease.
-

3 ☐ The cutting off of Indian rice imports due to the Japanese invasion of Burma

4 ☐ Panic buying by public due to the confiscation of the available rice in Bengal by Raj officials.



**Solution:**

**Correct Answer : 4**

**Your Answer : 4**

**Refer Para 3. The fear of a Japanese invasion led to the confiscation of everything that might help the Japanese including rice. This confiscation of rice created panic in the public and they began hoarding. So, 4 is the immediate cause. 2 and 3 did not trigger hoarding.**

 Answer key/Solution

Bookmark

FeedBack

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#### **Q.16 [11594329]**

Which of the following situations is the most similar to the one mentioned in the last sentence of Para 4 “The point was...price of rice.”?

1 ☐ Apple keeps on launching new variants of its iPhone in the market. As consumers became aware about the launch of its new variant, the price of its latest available variant fell. This latest variant was very similar to the one being launched.



2 ○ De Beers diamond cartel acquired almost all the diamond mines and charged an exorbitant price for its diamonds. Another small diamond cartel De Vera landed upon huge diamond reserves, and once this became public knowledge, the price of diamonds fell significantly in the market.

3 ○ Hearing the news of Coronavirus, some Indian traders, expecting a stock market crash and a rush to purchase a safe asset like gold, bought gold in huge quantities. Some days into the crash, the government discovered huge gold reserves. Once this was telecast on the news, the price of gold fell.

4 ○ The share price of Reliance Industries shot up on the National Stock Exchange (NSE), but not on the Bombay Stock Exchange (BSE). The knowledge of this price difference resulted in the investors buying shares from the BSE. Eventually, the price on the NSE fell from its previous level.



**Solution:**

**Correct Answer : 3**

**Your Answer : 3**

**1 is about different variants of a phone and does not have any element of hoarding. 2 is a situation of a dominant cartel. De Beers has acquired the source of supply; it hasn't hoarded diamonds from the market. So, it can be accused of profiteering, but not hoarding. 3 is a classic hoarding situation in the times of a crisis, where the knowledge of an alternative supply source (the gold reserves) made hoarding unprofitable. 4 again does not give reasons for the share price's shooting up, and so can't be termed as hoarding.**

 Answer key/Solution

Bookmark

FeedBack

**Direction for questions (15-18):** Read the given passage and answer the questions that follow.

Winston Churchill famously called Indians “a beastly people with a beastly religion who bred like rabbits”.

Madhushree Mukerjee’s Churchill’s Secret War establishes how Churchill and his associates could easily have stopped the Bengal famine of 1943 with a few shipments of food grains but refused, in spite of repeated appeals from two successive Viceroys, Churchill’s own Secretary of State for India and even the President of the United States.

Famines, never unknown in India, became increasingly lethal during the Raj because of the export of food grains and the replacement of food crops with indigo or jute. The Second World War made things worse, especially after Japanese forces occupied Burma in 1942, cutting off Indian rice imports. Then a destructive cyclone hit the Bengal coast just when the crucial winter crop was maturing, and the surviving rice was damaged by disease. Officials of the Raj, fearing a Japanese invasion, confiscated everything that might help the invading force – boats, carts, motor vehicles, elephants and, crucially, all the rice available. The Japanese never came but a panicking public – and many crafty businessmen – immediately began to hoard rice and the staple food of the people quickly disappeared from the marketplace.

Government stocks were released but only to feed the people of Calcutta and the urban population never suffered too greatly. The rural masses, however, were left to the wolves. This was when Churchill could have made a difference by sending wheat or rice to Bengal, and not enormous quantities. The point was to make hoarding unprofitable and as the Viceroy Lord Linlithgow pointed out, “the mere knowledge of impending imports” would have done so by lowering the price of rice.

Churchill and his war cabinet, however, decided to reserve available shipping to take food to Italy in case it fell to the Allies. He even refused help from Subhas Chandra Bose, then fighting with Axis forces, and also from Australia and Canada.

So, hundreds of thousands perished in the villages of Bengal. In London, Churchill’s beloved advisor, the physicist Frederick Alexander Lindemann (Lord Cherwell), was unmoved. A firm believer in Malthusian population theory, he blamed Indian philoprogenitiveness for the famine – sending more food would worsen the situation by encouraging Indians to breed more. The prime minister was of the same opinion and expressed himself so colourfully that Leo Amery, Secretary of State for India, exploded at him, comparing his attitudes to Hitler’s.

The Churchill industry has always denied that their idol could have done anything to relieve the Bengal famine. Shipping, they claim, was scarce and it just wasn’t possible to send food to Bengal. Mukerjee nails those “terminological inexactitudes” with precision. There was a shipping glut in summer and autumn 1943, thanks to the US transferring cargo ships to British control. Churchill, Lindemann and their close associates simply did not consider Indian lives worth saving. [...]

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**Q.17 [11594329]**

Based on the passage, one could best characterize Churchill as:

- 
- 1 ☐ An arrogant racist who was known to be inspired by Hitler’s attitudes.
- 
- 2 ☐ An obstinate leader, who was severely prejudiced and antipathetic towards Indians.
-

3 ☐ An inconsiderate leader who, because of his fear of the Japanese invasion, made Indians suffer greatly.

4 ☐ A stubborn bigot who cared more about his political interests than about the lives of his people.



**Solution:**

**Correct Answer : 2**

**Your Answer : 2**

**1 is incorrect as the passage states that "Leo Amery, Secretary of State for India, exploded at him, comparing his attitudes to Hitler's." That does not imply that Churchill was "known to be" inspired by Hitler's attitudes. 2 is correct because Para 2 shows his obstinacy, Para 1 and the last sentence of the passage show his prejudice, and his antipathy towards Indians.**

 Answer key/Solution

Bookmark

FeedBack

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Madhushree Mukerjee’s Churchill’s Secret War establishes how Churchill and his associates could easily have stopped the Bengal famine of 1943 with a few shipments of food grains but refused, in spite of repeated appeals from two successive Viceroys, Churchill’s own Secretary of State for India and even the President of the United States.

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**Q.18 [11594329]**

Based on the passage, which of the following is the Churchill industry most likely to state to support their assertion that Churchill could not have done anything to relieve the famine?

- 1 ☐ Even if there weren’t a shortage of ships during 1943, sending food shipments to India would have certainly resulted in Churchill’s losing popular support in Britain.
-

2 ○ In the face of a shipping scarcity, Churchill ordered the British government in India to release vast quantities of food-grains and distribute them to rural areas as well, but that couldn't happen due to very poor accessibility to these areas.

3 ○ Even if there weren't a shipping shortage and Churchill had sent food-grains to India, the Japanese would have definitely attacked India and Bengal would have been the first one to be enslaved by them.

4 ○ If it had not been for Churchill's efforts, the casualties of the famine would have been far more than they turned out to be.



**Solution:**

**Correct Answer : 2**

**Your Answer : 2**

Answer key/Solution

**1 talks about Churchill giving priority to his political interests and not an inability to relieve the famine. So, it won't strengthen the argument. 3 talks about another difficulty that could have arisen as a result of Churchill's efforts, and not his inability to relieve the famine. 4 is just a comment that has not been substantiated with reasons. No details about his efforts and how they helped have been given. 2 supports the assertion by stating that despite a shipping shortage, Churchill ordered the distribution of vast quantities. They couldn't reach rural areas because of extremely poor accessibility, and not because of a lack of his efforts. He couldn't have done much beyond it.**

Bookmark

FeedBack

### Q.19 [11594329]

**Directions for question (19):** The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. The aphorism both resists and invites fuller articulations, and so hermeneutics – a conduit between the text and the reader – is demanded from every reader.
2. The nature of Heraclitus' microform is true for the aphorism in general.
3. They are antithetical because they are too enigmatic, and propaedeutic because they cannot but prompt us to think about the origins and nature of things.
4. Fragments are both *antithetical* and *propaedeutic* to argumentative philosophy.



**Solution:****Correct Answer : 2143****Your Answer : 4231**[🔍 Answer key/Solution](#)

Though this paragraph appears difficult because of the esoteric terms that have been used in it, we can easily see a link between 3 and 4, such that 3 follows 4. Also, the fragments being talked about are fragments of aphorism. So, neither 3 nor 4 can be the opening sentence. 1 is closely related with 4 and 3. Resisting fuller articulations is linked to "enigmatic" in 3 and inviting fuller articulation is linked to "prompt us to think about the origins and nature of things. 2 introduces the discussion by saying that the nature of microform (a miniaturized reproduction) fits an aphorism, and then 1 elaborates on it. So, 2143.

[Bookmark](#)[FeedBack](#)**Q.20 [11594329]**

**Directions for question (20):** The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. Until 200 years ago the notion of loneliness barely existed.
2. In 2018, the first British Minister for Loneliness was appointed, and the government published *A Connected Society*.
3. The histories detailed in Vincent and Bound Alberti's books reveal what a striking development this is.
4. In the preface, the country's then prime minister Theresa May referred to research that "shows that loneliness is as damaging to our physical health as smoking".

[✕](#)**Solution:****Correct Answer : 2431****Your Answer : 4312**[🔍 Answer key/Solution](#)

While one may be tempted to think of 1 and 2 as a mandatory pair, 1 fits better with 3, because it shows what the "histories reveal" and what makes the 2018 development "striking". 2 and 4 is an easy mandatory pair. So, the sequence is 2431.

[Bookmark](#)[FeedBack](#)

**Q.21 [11594329]**

**Directions for question (21):** The passage given below is followed by four summaries. Choose the option that best captures the author's position.

Each of the sciences concerns itself with a particular set of things that exist – and nothing more. And here's where things become highly disputable. Heidegger asks: But what about this 'nothing'? Is it only an accident we speak like that quite naturally? Is it only a manner of speaking – and nothing more?

This slightly mad query launches Heidegger into a discussion of the Nothing. 'If science is right then one thing stands firm: science wishes to know nothing of Nothing,' he writes. That is, the objects of scientific enquiry, whatever they are, are the only things that the scientist is interested in – and nothing else. Heidegger is playing on this phrase 'nothing else' and reifying it into the Nothing.

- 1 ☐ Heidegger questions the nature of 'nothing' that the scientists are not interested in and relates it to the larger concept of Nothing in order to make it more concrete.
- 2 ☐ Heidegger relates the phrase 'nothing more' in the fact that scientists are interested only in the things that exist and nothing more to the Nothing and asserts that they are not interested in knowing about the Nothing.
- 3 ☐ Because scientists are concerned only about the things that exist and 'nothing more', Heidegger reifies this phrase to the 'Nothing' by stating that they are not interested in knowing about the things that do not exist.
- 4 ☐ Heidegger indulges in wordplay when he picks up the phrase 'nothing more' from how the scope of each of the sciences is defined and relates it to his discussion of the Nothing.

**Solution:**

**Correct Answer : 2**

 Answer key/Solution

**1 is rejected because Heidegger does not question the nature of 'nothing', in fact the nature of 'nothing' is not known. Also, using the words 'the larger concept' for Nothing is in the nature of a conclusion, not a summary. 2 captures the relationship of 'nothing' to 'Nothing' established by Heidegger aptly. 3 is incorrect because Nothing has not been defined in the paragraph. It can't be called 'things that do not exist'. This option is alien to the para. Heidegger's indulging in wordplay is a part of the para, not the essence of it. It is narrow.**

Bookmark

FeedBack

**Q.22 [11594329]**

**Directions for question (22):** Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

1. But there is no doubt that it often compounds the confusion as the ghosts of the past are made to walk again in a landscape that has changed profoundly.
2. For all its other virtues, comparison does not do well with the novelty that Trump certainly represents, for all of his preconditions and sources.
3. It is true that in the face of novelty, analogy with possible historical avatars is indispensable, to abate confusion and to seek orientation.
4. Yet people forget that analogy had commonly seemed noxious, not necessary, in the previous century.
5. It is no contradiction that comparing our current situation in America to fascism spares ourselves the trouble of analyzing what is really new about it.



**Solution:**

**Correct Answer : 4**

**Your Answer : 5**

 Answer key/Solution

The para talks about how comparisons are not always good. 5 and 2 form a pair as they talk about the current situation in America. 3 and 1 form a pair because while 3 talks about a merit of comparisons, 1 contradicts it with a demerit. 4 just places comparisons in context by saying how they seemed harmful in the previous century. So, it is the odd one out.

Bookmark

FeedBack

**Q.23 [11594329]**

**Directions for question (23):** Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

1. On the other side of the Atlantic, the American Civil War saw several outbreaks among young fighting men even though they technically had never left their homeland, per se.
2. To stop the spread of the disease, Russian physicians recommended burying alive anyone who started showing symptoms—which apparently did prove quite effective.
3. Their physicians were a bit kinder, suggesting that occasional removal from front line fighting would bolster their spirits.
4. As it moved out of the medical realm and into the cultural, though, nostalgia did not fully shed its strange stigma.
5. By the 1800s, the terrors of nostalgia finally spread to various countries' soldiers.



**Solution:****Correct Answer : 4**[Answer key/Solution](#)

The passage talks about nostalgia as a disease. The sequence is 5213. 4 talks about nostalgia moving into the cultural realm and does not match the scope of the discussion.

[Bookmark](#)[FeedBack](#)**Q.24 [11594329]**

**Directions for question (24):** The passage given below is followed by four summaries. Choose the option that best captures the author's position.

In the New Yorker's schema, "Night Life" does not refer to clubs and parties (of little interest, apparently, to the magazine's readership) or bars (covered under "Food & Drink"). Instead, the heading serves as a euphemism for non-classical music. In the past, a classical music/nightlife dichotomy may have reflected a broader social consensus on the distinction between high and low forms. But with such distinctions murky if not altogether irrelevant today, it is hard to ignore the inherent racism: in practice, the New Yorker maintains one category for a historically white kind of music, and another category for everything else.

- 1 ☐ The way the magazine New Yorker covers all non-white music under the label "Night Life" and all white music under "Classical Music" is racist.
- 2 ☐ While New Yorker's distinction between classical music/nightlife may have reflected a consensus in the past, covering non-classical music under the label "Night Life" today is murky in the least and racist at the worst.
- 3 ☐ The way New Yorker covers all historically white kind of music—classical—in one category and other music under the label "Nightlife" might have reflected a consensus in the past, but its racism is difficult to ignore.
- 4 ☐ The New Yorker's murky practice of calling white music "Classical music" and non-white music as "Night life" has always been racist and is irrelevant today.

**Solution:****Correct Answer : 3**[Answer key/Solution](#)

1 is rejected as it does not comprehensively cover the historical aspect of the dichotomy. 2 is rejected because it says that 'today' the practice is racist 'at the worst'. The author outrightly calls the practice inherently racist, so it has always been racist. 4 is rejected because it says that the practice is irrelevant today, but the author acknowledges that the might not be altogether irrelevant today in the phrase 'if not altogether irrelevant' today. Also, it doesn't cover the past scenario. 3 captures the essence.

[Bookmark](#)[FeedBack](#)

**Q.25 [11594329]**

**Directions for question (25):** The passage given below is followed by four summaries. Choose the option that best captures the author's position.

As I sit in a communal garden in London at sunset and listen to more familiar, supposedly less exotic, birds, I recall Hudson's *Birds in London*. In one of its essays on sparrows, which for Hudson stood for wild nature in an urban wasteland, he wrote, "it is always possible to find something fresh to say of a bird of so versatile a mind". When I first read this, I was surprised not only by that word "fresh" but by the notion that a sparrow has a "mind". Hudson praises this humble little bird for "its greater intelligence" and "individual character". He found that, despite their ubiquity, "the individual sparrow is little known to us". Here was Hudson looking at a common bird as if for the first time.

- 1 ☐ Hudson thought that sparrows, little known creatures of greater intelligence and individual character, always provided one with something fresh to say about them.
- 2 ☐ The author recalls how Hudson, in his 'Birds in London' looks at a sparrow as if for the first time and calls it a little-known creature of greater intelligence and individual character about whom one could always say something fresh.
- 3 ☐ The author, while citing some instances from an essay in Hudson's 'Birds in London', states how Hudson could look at a common bird—sparrow—as if for the time, and admits to being surprised when he first read some of his words.
- 4 ☐ The author admires Hudson's ability to look at a sparrow as if for the first time but is surprised at his notion that sparrows have a "mind" and one can always say something "fresh" about them.

**Solution:**

**Correct Answer : 3**

 **Answer key/Solution**

**1 does not say anything about the author, just about Hudson. 3 is more comprehensive than 2 as it captures the author's surprise when he first read some of Hudson's words. 4 is distorted as the author was surprised "at first". 4 implies that he doubts what Hudson said about the sparrow.**

Bookmark

FeedBack

## Sec 2

**Directions for questions 26 to 28:** Answer the questions on the basis of the information given below.

A dance group CRAZY CREW of college NIFF, Gujarat participated in a dance competition in an annual fest of XYZ college.

According to the rules of the competition, the group had to perform four types of dances D1, D2, D3, and D4, not necessarily in that order. Each type of dance had same number of dancers but the arrangement of these dancers on the stage during a particular dance type can be different. The arrangement of dancers on the stage for the different dance types is as follows:

D1 : All dancers should perform in an arrangement the structure of which has multiple rows facing the judges such that number of dancers in  $n$ th row ( $R_n$ ) is  $n$ .

D2 : All dancers should dance in an arrangement the structure of which is a  $n \times n$  grid, having same number of dancers in each row as well as in each column.

D3 : All dancers should dance in an arrangement having one or more rows with exactly 7 dancers in each row.

D4 : All dancers should dance in an arrangement having one or more rows with exactly 9 dancers in each row.

Since the number of dancers in each type of dances of the dance group has to be same, the team-lead of the group was not able to decide which dance type to perform in which order, and how many dancers to include in his group. Moreover, the judges revised the rule of performing all type of dances and asked him to perform at least 3 types of dances in any sequence.

#### Q.26 [11594329]

What is the minimum number of dancers required to perform at least 3 types of dances in any sequence?

**Solution:**

**Correct Answer : 36**

The possible number of dancers required for each type of the dance is as given below.

D1 : 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105, ...

D2 : 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, ...

D3 : 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, 91, 98, ...

D4 : 9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, ...

The group can perform D1, D2 and D4 in any sequence with minimum 36 dancers.

Bookmark

FeedBack

 Answer key/Solution

**Directions for questions 26 to 28:** Answer the questions on the basis of the information given below.

A dance group CRAZY CREW of college NIFF, Gujarat participated in a dance competition in an annual fest of XYZ college.

According to the rules of the competition, the group had to perform four types of dances D1, D2, D3, and D4, not necessarily in that order. Each type of dance had same number of dancers but the arrangement of these dancers on the stage during a particular dance type can be different. The arrangement of dancers on the stage for the different dance types is as follows:

D1 : All dancers should perform in an arrangement the structure of which has multiple rows facing the judges such that number of dancers in  $n$ th row ( $n$ ) is  $n$ .

D2 : All dancers should dance in an arrangement the structure of which is a  $n \times n$  grid, having same number of dancers in each row as well as in each column.

D3 : All dancers should dance in an arrangement having one or more rows with exactly 7 dancers in each row.

D4 : All dancers should dance in an arrangement having one or more rows with exactly 9 dancers in each row.

Since the number of dancers in each type of dances of the dance group has to be same, the team-lead of the group was not able to decide which dance type to perform in which order, and how many dancers to include in his group. Moreover, the judges revised the rule of performing all type of dances and asked him to perform at least 3 types of dances in any sequence.

**Q.27 [11594329]**

If in D1 there is no row having both male and female, and also no two consecutive rows can have dancers of the same gender, what is the minimum number of female dancers required to perform at least three types of dances with D1 as one of it?

1 ☐ 12

2 ☐ 14

3 ☐ 16

4 ☐ 21

**Solution:**

**Correct Answer : 3**

 Answer key/Solution

The possible number of dancers required for each type of the dance is as given below.

D1 : 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105, ...

D2 : 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, ...

D3 : 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, 91, 98, ...

D4 : 9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, ...

In D1 the minimum number of female dancers required to perform at least three dances =  $1 + 3 + 5 + 7 = 16$ .

Bookmark

FeedBack

**Directions for questions 26 to 28:** Answer the questions on the basis of the information given below.

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D1 : All dancers should perform in an arrangement the structure of which has multiple rows facing the judges such that number of dancers in  $n$ th row ( $R_n$ ) is  $n$ .

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Since the number of dancers in each type of dances of the dance group has to be same, the team-lead of the group was not able to decide which dance type to perform in which order, and how many dancers to include in his group. Moreover, the judges revised the rule of performing all type of dances and asked him to perform at least 3 types of dances in any sequence.

#### Q.28 [11594329]

If the judges allow the team-lead to increase or decrease the number of dancers by 1 only once, what is the minimum number of dancers required to start with to perform at least 3 types of dances?

1 ☐ 7

2 ☐ 9

3 ☐ 12

4 ☐ 14

**Solution:**

**Correct Answer : 2**

 Answer key/Solution

The possible number of dancers required for each type of the dance is as given below.

D1 : 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105, ...

D2 : 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, ...

D3 : 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, 91, 98, ...

D4 : 9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, ...

With 9 dancers, one can perform, D2, D4 and D1 (by reducing one dancer in D1 after performing D2 and D4).

Bookmark

FeedBack

**Directions for questions 29 to 31:** Answer the questions on the basis of the information given below.

Priyanshu recently visited Casino and played darts game for six days in a week. There were three circles on the dart board. Red the inner circle, yellow the middle circle and green the outer circle. For hitting the red circle he scored 7 points, for hitting the yellow circle he scored 2 points and if he hit the green circle, there was a penalty of 5 points. On one of the six days he tried it for less than 10 times and for the rest of days he tried it more than 10 times but not more than 18 times.

1. The number of times he hit the red circle was distinct for each day.
2. The amount he earned by scoring points is a multiple of 100 and also he had to pay the penalty points in multiple of 100s only.
3. He hit the red circle at least once on each day.

The following tables give the information about the number of times he hit the green circle on all the six days and the overall amount of money, in Rs., (earned and paid as a penalty) he had at the end of the day after he played the dart game.

| Day | Green Penalty stroke |
|-----|----------------------|
| 1   | 3                    |
| 2   | 4                    |
| 3   | 1                    |
| 4   | 2                    |
| 5   | 6                    |
| 6   | 8                    |

| Day | Amount received/paid |
|-----|----------------------|
| 1   | 3300                 |
| 2   | 4600                 |
| 3   | 7300                 |
| 4   | 2000                 |
| 5   | 2100                 |
| 6   | -2200                |

**Q.29 [11594329]**

How many total number of times he hit the red circle for all 6 days together?

1 ☐ 37

2 ☐ 35

3 ☐ 47

4 ☐ Cannot be determined



**Solution:****Correct Answer : 1****Your Answer : 1** Answer key/Solution

All amounts given in table 2 are multiple of 100. On day 1, the amount earned is Rs. 3,300 so that means the total points (including penalty points) on that day will be 33.

Further there are 7 points for hitting the red circle, 2 points for hitting the yellow circle and 5 penalty points for hitting the green circle.

Let  $x$ ,  $y$  and  $z$  be the number of times he hit the red, yellow and green circles respectively. So, for day 1, we can form an equation,  $7x + 2y - 5z = 33$ . As we also know the number of times the green circle is hit and total points scored for each day, so we can have six equations to form.

Day 1.  $7x + 2y - 15 = 33$

$$7x + 2y = 48$$

$$\left. \begin{array}{l} x = 4 \quad y = 10 \\ x = 6 \quad y = 3 \end{array} \right\} \text{Possible cases}$$

Day 2.  $7x + 2y - 20 = 46$

On solving the possible values of  $x$  and  $y$  could be

$$\left. \begin{array}{l} x = 2 \quad y = 26 \\ x = 4 \quad y = 19 \\ x = 6 \quad y = 12 \end{array} \right\} \text{more than 18 hits}$$

$$x = 8 \quad y = 5 \} \text{Only this case satisfies}$$

Day 3.  $7x + 2y - 5 = 73$

Only case that satisfies is when  $x = 10$  and  $y = 4$ .

Day 4.  $7x + 2y - 10 = 20$

Two cases are possible.

$$\left. \begin{array}{l} x = 2 \quad y = 8 \\ x = 4 \quad y = 1 \end{array} \right\}$$

Day 5.  $7x + 2y - 30 = 21$

$$7x + 2y = 51$$

One case is possible only

$$x = 7 \quad y = 1$$

Day 6.  $7x + 2y - 40 = -22$

$$7x + 2y = 18$$

Only one case is possible.

$$x = 2 \quad y = 2$$

Out of these six days, only day 1 and day 4 have two possible cases. As given in the question, on one of the six days the total number of trials has to be less than 10 so, on 4th day, that's possible so we will consider that case only when  $x = 4$  and  $y = 1$  so, now on day 1,  $x$  cannot be 4 as it is given that  $x$  has to be distinct for each day.

So, final table will look like.

| Day | 1 | 2 | 3  | 4 | 5 | 6 |
|-----|---|---|----|---|---|---|
| X   | 6 | 8 | 10 | 4 | 7 | 2 |
| Y   | 3 | 5 | 4  | 1 | 1 | 2 |
| Z   | 3 | 4 | 1  | 2 | 6 | 8 |

The total number of times he hit the red circle for all 6 days together is 37.

Bookmark

FeedBack

**Directions for questions 29 to 31:** Answer the questions on the basis of the information given below.

Priyanshu recently visited Casino and played darts game for six days in a week. There were three circles on the dart board. Red the inner circle, yellow the middle circle and green the outer circle. For hitting the red circle he scored 7 points, for hitting the yellow circle he scored 2 points and if he hit the green circle, there was a penalty of 5 points. On one of the six days he tried it for less than 10 times and for the rest of days he tried it more than 10 times but not more than 18 times.

1. The number of times he hit the red circle was distinct for each day.
2. The amount he earned by scoring points is a multiple of 100 and also he had to pay the penalty points in multiple of 100s only.
3. He hit the red circle at least once on each day.

The following tables give the information about the number of times he hit the green circle on all the six days and the overall amount of money, in Rs., (earned and paid as a penalty) he had at the end of the day after he played the dart game.

| Day | Green Penalty stroke |
|-----|----------------------|
| 1   | 3                    |
| 2   | 4                    |
| 3   | 1                    |
| 4   | 2                    |
| 5   | 6                    |
| 6   | 8                    |

| Day | Amount received/paid |
|-----|----------------------|
| 1   | 3300                 |
| 2   | 4600                 |
| 3   | 7300                 |
| 4   | 2000                 |
| 5   | 2100                 |
| 6   | -2200                |

**Q.30 [11594329]**

On which day Priyanshu hit the yellow circle for the maximum number of times?

1 ☐ Day 1

2 ☐ Day 2

3 ☐ Day 5

4 ☐ Cannot be determined





**Solution:****Correct Answer : 2****Your Answer : 2** Answer key/Solution

All amounts given in table 2 are multiple of 100. On day 1, the amount earned is Rs. 3,300 so that means the total points (including penalty points) on that day will be 33.

Further there are 7 points for hitting the red circle, 2 points for hitting the yellow circle and 5 penalty points for hitting the green circle.

Let  $x$ ,  $y$  and  $z$  be the number of times he hit the red, yellow and green circles respectively. So, for day 1, we can form an equation,  $7x + 2y - 5z = 33$ . As we also know the number of times the green circle is hit and total points scored for each day, so we can have six equations to form.

Day 1.  $7x + 2y - 15 = 33$

$$7x + 2y = 48$$

$$\left. \begin{array}{l} x = 4 \quad y = 10 \\ x = 6 \quad y = 3 \end{array} \right\} \text{Possible cases}$$

Day 2.  $7x + 2y - 20 = 46$

On solving the possible values of  $x$  and  $y$  could be

$$\left. \begin{array}{l} x = 2 \quad y = 26 \\ x = 4 \quad y = 19 \\ x = 6 \quad y = 12 \end{array} \right\} \text{more than 18 hits}$$

$$x = 8 \quad y = 5 \} \text{Only this case satisfies}$$

Day 3.  $7x + 2y - 5 = 73$

Only case that satisfies is when  $x = 10$  and  $y = 4$ .

Day 4.  $7x + 2y - 10 = 20$

Two cases are possible.

$$\left. \begin{array}{l} x = 2 \quad y = 8 \\ x = 4 \quad y = 1 \end{array} \right\}$$

Day 5.  $7x + 2y - 30 = 21$

$$7x + 2y = 51$$

One case is possible only

$$x = 7 \quad y = 1$$

Day 6.  $7x + 2y - 40 = -22$

$$7x + 2y = 18$$

Only one case is possible.

$$x = 2 \quad y = 2$$

Out of these six days, only day 1 and day 4 have two possible cases. As given in the question, on one of the six days the total number of trials has to be less than 10 so, on 4th day, that's possible so we will consider that case only when  $x = 4$  and  $y = 1$  so, now on day 1,  $x$  cannot be 4 as it is given that  $x$  has to be distinct for each day.

So, final table will look like.

| Day | 1 | 2 | 3  | 4 | 5 | 6 |
|-----|---|---|----|---|---|---|
| X   | 6 | 8 | 10 | 4 | 7 | 2 |
| Y   | 3 | 5 | 4  | 1 | 1 | 2 |
| Z   | 3 | 4 | 1  | 2 | 6 | 8 |

On day 2, he hit the yellow circle the maximum number of times.

Bookmark

FeedBack

**Directions for questions 29 to 31:** Answer the questions on the basis of the information given below.

Priyanshu recently visited Casino and played darts game for six days in a week. There were three circles on the dart board. Red the inner circle, yellow the middle circle and green the outer circle. For hitting the red circle he scored 7 points, for hitting the yellow circle he scored 2 points and if he hit the green circle, there was a penalty of 5 points. On one of the six days he tried it for less than 10 times and for the rest of days he tried it more than 10 times but not more than 18 times.

1. The number of times he hit the red circle was distinct for each day.
2. The amount he earned by scoring points is a multiple of 100 and also he had to pay the penalty points in multiple of 100s only.
3. He hit the red circle at least once on each day.

The following tables give the information about the number of times he hit the green circle on all the six days and the overall amount of money, in Rs., (earned and paid as a penalty) he had at the end of the day after he played the dart game.

| Day | Green Penalty stroke |
|-----|----------------------|
| 1   | 3                    |
| 2   | 4                    |
| 3   | 1                    |
| 4   | 2                    |
| 5   | 6                    |
| 6   | 8                    |

| Day | Amount received/paid |
|-----|----------------------|
| 1   | 3300                 |
| 2   | 4600                 |
| 3   | 7300                 |
| 4   | 2000                 |
| 5   | 2100                 |
| 6   | -2200                |

**Q.31 [11594329]**

For how many days it is possible to uniquely determine the number of hits on red circle and yellow circle?

1 ☐ 4 days

2 ☐ 5 days

3 ☐ 6 days

4 ☐ 2 days



**Solution:****Correct Answer : 3****Your Answer : 3** Answer key/Solution

All amounts given in table 2 are multiple of 100. On day 1, the amount earned is Rs. 3,300 so that means the total points (including penalty points) on that day will be 33.

Further there are 7 points for hitting the red circle, 2 points for hitting the yellow circle and 5 penalty points for hitting the green circle.

Let  $x$ ,  $y$  and  $z$  be the number of times he hit the red, yellow and green circles respectively. So, for day 1, we can form an equation,  $7x + 2y - 5z = 33$ . As we also know the number of times the green circle is hit and total points scored for each day, so we can have six equations to form.

Day 1.  $7x + 2y - 15 = 33$

$$7x + 2y = 48$$

$$\left. \begin{array}{l} x = 4 \quad y = 10 \\ x = 6 \quad y = 3 \end{array} \right\} \text{Possible cases}$$

Day 2.  $7x + 2y - 20 = 46$

On solving the possible values of  $x$  and  $y$  could be

$$\left. \begin{array}{l} x = 2 \quad y = 26 \\ x = 4 \quad y = 19 \\ x = 6 \quad y = 12 \end{array} \right\} \text{more than 18 hits}$$

$$x = 8 \quad y = 5 \} \text{Only this case satisfies}$$

Day 3.  $7x + 2y - 5 = 73$

Only case that satisfies is when  $x = 10$  and  $y = 4$ .

Day 4.  $7x + 2y - 10 = 20$

Two cases are possible.

$$\left. \begin{array}{l} x = 2 \quad y = 8 \\ x = 4 \quad y = 1 \end{array} \right\}$$

Day 5.  $7x + 2y - 30 = 21$

$$7x + 2y = 51$$

One case is possible only

$$x = 7 \quad y = 1$$

Day 6.  $7x + 2y - 40 = -22$

$$7x + 2y = 18$$

Only one case is possible.

$$x = 2 \quad y = 2$$

Out of these six days, only day 1 and day 4 have two possible cases. As given in the question, on one of the six days the total number of trials has to be less than 10 so, on 4th day, that's possible so we will consider that case only when  $x = 4$  and  $y = 1$  so, now on day 1,  $x$  cannot be 4 as it is given that  $x$  has to be distinct for each day.

So, final table will look like.

| Day | 1 | 2 | 3  | 4 | 5 | 6 |
|-----|---|---|----|---|---|---|
| X   | 6 | 8 | 10 | 4 | 7 | 2 |
| Y   | 3 | 5 | 4  | 1 | 1 | 2 |
| Z   | 3 | 4 | 1  | 2 | 6 | 8 |

For all 6 days, it is possible to uniquely determine the number of hits on red circle and yellow circle.

Bookmark

FeedBack

**Directions for questions 32 to 35:** Answer the questions on the basis of the information given below.

Six friends - A, B, C, D, E, and F - participate in an online quiz having 4 rounds. In every round, participants are awarded points and the top three participants of each round get prizes with 1st prize being a LED-TV, 2nd being a Refrigerator and 3rd being a cash prize of Rs. 10,000. All six friends participate in every round of the quiz.

Some details about who-won-what in every round is given below:

- Cash prize is won by only 3 participants and LED-TV is won by a different participant in each round.
- A wins in exactly three rounds and receives two types of prizes. C wins a Refrigerator only.
- There is only one participant who wins a prize in every round but never comes 3rd in any round.
- E wins a prize in two rounds only by coming 1st and 3rd. D wins a LED-TV only.
- D and E never win a prize in the same round. A and C win in the same round.

**Q.32 [11594329]**

How many prizes does F win in the online quiz?

1 ☐ 1

2 ☐ 2

3 ☐ 4

4 ☐ Either (1) or (3)

**Solution:**

**Correct Answer : 4**

 Answer key/Solution

It is very clear that a total of 12 prizes are distributed in 4 rounds, with 4 each of LED-TV (L), Refrigerator(R) and cash prize of Rs. 10,000(C). Now 'A' has won 2 types of prizes in 3 rounds, this means 'A' has won one of the prizes twice. C and D have won a single prize that is a refrigerator and a LED-TV respectively. Also, E has won two prizes - LED-TV and Cash Prize. This leaves us with two possibilities that either B or F has won 4 prizes.

**Case I:** Let B be the one who won 4 prizes; therefore, F will win one prize only.

We know that LED-TV is won by a different participant in each round, therefore, B must win one LED-TV and 3 – refrigerators. And Now, A won 2 types of prizes, so they must be 1- LED-TV and 2 – cash prizes.

| Friend | Prize      |
|--------|------------|
| A (3)  | L, C, C    |
| B (4)  | L, R, R, R |
| C (1)  | R          |
| D (1)  | L          |
| E (2)  | L, C       |
| F (1)  | C          |

**Case II:** Let F be the one who won 4 prizes; therefore, B will win one prize only.

We know that LED-TV is won by a different participant in each round, therefore, F must win one LED-TV and 3 – refrigerators. And Now, A won 2 types of prizes, so they must be 1- LED-TV and 2 – cash prizes.

| Friend | Prize   |
|--------|---------|
| A (3)  | L, C, C |
| B (1)  | C       |

|       |            |
|-------|------------|
| C (1) | R          |
| D (1) | L          |
| E (2) | L, C       |
| F (4) | L, R, R, R |

**Note:** In above mentioned cases the prizes are won in any order of rounds.

We can make cases for Rounds as well-

**Case I:** B wins 4 prizes.

Let us assume that A, B, D and E won LED-TV in round 1, 2, 3 and 4 respectively. Now since B won first prize in round 2, therefore, second prize must be won in round 1, 3 and 4. And therefore, C won refrigerator in round 2 and since A win in the same round as C, therefore, C won cash prize in round 2. And because D and E never wins a prize in the same round therefore, E must win in round 1.

|         | 1st Prize (L) | 2nd Prize (R) | 3rd Prize (C) |
|---------|---------------|---------------|---------------|
| Round 1 | A             | B             | E             |
| Round 2 | B             | C             | A             |
| Round 3 | D             | B             | F or A        |
| Round 4 | E             | B             | A or F        |

**Case II:** F wins 4 prizes.

In this case everything remains same except B and F will interchange places.

|         | 1st Prize (L) | 2nd Prize (R) | 3rd Prize (C) |
|---------|---------------|---------------|---------------|
| Round 1 | A             | F             | E             |
| Round 2 | F             | C             | A             |
| Round 3 | D             | F             | B or A        |
| Round 4 | E             | F             | A or B        |

F won either 1 or 4 prizes in the Online Quiz.

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**Directions for questions 32 to 35:** Answer the questions on the basis of the information given below.

Six friends - A, B, C, D, E, and F - participate in an online quiz having 4 rounds. In every round, participants are awarded points and the top three participants of each round get prizes with 1st prize being a LED-TV, 2nd being a Refrigerator and 3rd being a cash prize of Rs. 10,000. All six friends participate in every round of the quiz.

Some details about who-won-what in every round is given below:

- Cash prize is won by only 3 participants and LED-TV is won by a different participant in each round.
- A wins in exactly three rounds and receives two types of prizes. C wins a Refrigerator only.
- There is only one participant who wins a prize in every round but never comes 3rd in any round.
- E wins a prize in two rounds only by coming 1st and 3rd. D wins a LED-TV only.
- D and E never win a prize in the same round. A and C win in the same round.

### Q.33 [11594329]

Who among the following necessarily wins a cash prize?

1 ○ B

2 ○ F

3 ○ A

4 ○ C

**Solution:****Correct Answer : 3** Answer key/Solution

It is very clear that a total of 12 prizes are distributed in 4 rounds, with 4 each of LED-TV (L), Refrigerator(R) and cash prize of Rs. 10,000(C). Now 'A' has won 2 types of prizes in 3 rounds, this means 'A' has won one of the prizes twice. C and D have won a single prize that is a refrigerator and a LED-TV respectively. Also, E has won two prizes - LED-TV and Cash Prize. This leaves us with two possibilities that either B or F has won 4 prizes.

**Case I: Let B be the one who won 4 prizes; therefore, F will win one prize only.**

We know that LED-TV is won by a different participant in each round, therefore, B must win one LED-TV and 3 – refrigerators. And Now, A won 2 types of prizes, so they must be 1- LED-TV and 2 – cash prizes.

| Friend | Prize      |
|--------|------------|
| A (3)  | L, C, C    |
| B (4)  | L, R, R, R |
| C (1)  | R          |
| D (1)  | L          |
| E (2)  | L, C       |
| F (1)  | C          |

**Case II: Let F be the one who won 4 prizes; therefore, B will win one prize only.**

We know that LED-TV is won by a different participant in each round, therefore, F must win one LED-TV and 3 – refrigerators. And Now, A won 2 types of prizes, so they must be 1- LED-TV and 2 – cash prizes.

| Friend | Prize      |
|--------|------------|
| A (3)  | L, C, C    |
| B (1)  | C          |
| C (1)  | R          |
| D (1)  | L          |
| E (2)  | L, C       |
| F (4)  | L, R, R, R |

**Note:** In above mentioned cases the prizes are won in any order of rounds.

We can make cases for Rounds as well-

**Case I: B wins 4 prizes.**

Let us assume that A, B, D and E won LED-TV in round 1, 2, 3 and 4 respectively. Now since B won first prize in round 2, therefore, second prize must be won in round 1, 3 and 4. And therefore, C won refrigerator in round 2 and since A win in the same round as C, therefore, C won cash prize in round 2. And because D and E never wins a prize in the same round therefore, E must win in round 1.

|         | 1st Prize (L) | 2nd Prize (R) | 3rd Prize (C) |
|---------|---------------|---------------|---------------|
| Round 1 | A             | B             | E             |
| Round 2 | B             | C             | A             |



|         |   |   |        |
|---------|---|---|--------|
| Round 3 | D | B | F or A |
| Round 4 | E | B | A or F |

Case II: F wins 4 prizes.

In this case everything remains same except B and F will interchange places.

|         | 1st Prize (L) | 2nd Prize (R) | 3rd Prize (C) |
|---------|---------------|---------------|---------------|
| Round 1 | A             | F             | E             |
| Round 2 | F             | C             | A             |
| Round 3 | D             | F             | B or A        |
| Round 4 | E             | F             | A or B        |

Participant A necessarily won a Cash Prize.

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**Directions for questions 32 to 35:** Answer the questions on the basis of the information given below.

Six friends - A, B, C, D, E, and F - participate in an online quiz having 4 rounds. In every round, participants are awarded points and the top three participants of each round get prizes with 1st prize being a LED-TV, 2nd being a Refrigerator and 3rd being a cash prize of Rs. 10,000. All six friends participate in every round of the quiz.

Some details about who-won-what in every round is given below:

- Cash prize is won by only 3 participants and LED-TV is won by a different participant in each round.
- A wins in exactly three rounds and receives two types of prizes. C wins a Refrigerator only.
- There is only one participant who wins a prize in every round but never comes 3rd in any round.
- E wins a prize in two rounds only by coming 1st and 3rd. D wins a LED-TV only.
- D and E never win a prize in the same round. A and C win in the same round.

#### Q.34 [11594329]

Who among the following necessarily wins in the same round with D?

1 ☐ A

2 ☐ B

3 ☐ C

4 ☐ Cannot be determined.

**Solution:**

**Correct Answer : 4**

[Answer key/Solution](#)

It is very clear that a total of 12 prizes are distributed in 4 rounds, with 4 each of LED-TV (L), Refrigerator(R) and cash prize of Rs. 10,000(C). Now 'A' has won 2 types of prizes in 3 rounds, this means 'A' has won one of the prizes twice. C and D have won a single prize that is a refrigerator and a LED-TV respectively. Also, E has won two prizes - LED-TV and Cash Prize. This leaves us with two possibilities that either B or F has won 4 prizes.

Case I: Let B be the one who won 4 prizes; therefore, F will win one prize only.

We know that LED-TV is won by a different participant in each round, therefore, B must win one LED-TV and 3 – refrigerators. And Now, A won 2 types of prizes, so they must be 1- LED-TV and 2 – cash prizes.

| Friend | Prize      |
|--------|------------|
| A (3)  | L, C, C    |
| B (4)  | L, R, R, R |
| C (1)  | R          |
| D (1)  | L          |
| E (2)  | L, C       |
| F (1)  | C          |

**Case II: Let F be the one who won 4 prizes; therefore, B will win one prize only.**

We know that LED-TV is won by a different participant in each round, therefore, F must win one LED-TV and 3 – refrigerators. And Now, A won 2 types of prizes, so they must be 1- LED-TV and 2 – cash prizes.

| Friend | Prize      |
|--------|------------|
| A (3)  | L, C, C    |
| B (1)  | C          |
| C (1)  | R          |
| D (1)  | L          |
| E (2)  | L, C       |
| F (4)  | L, R, R, R |

**Note:** In above mentioned cases the prizes are won in any order of rounds.

We can make cases for Rounds as well-

**Case I: B wins 4 prizes.**

Let us assume that A, B, D and E won LED-TV in round 1, 2, 3 and 4 respectively. Now since B won first prize in round 2, therefore, second prize must be won in round 1, 3 and 4. And therefore, C won refrigerator in round 2 and since A win in the same round as C, therefore, C won cash prize in round 2. And because D and E never wins a prize in the same round therefore, E must win in round 1.

|         | 1st Prize (L) | 2nd Prize (R) | 3rd Prize (C) |
|---------|---------------|---------------|---------------|
| Round 1 | A             | B             | E             |
| Round 2 | B             | C             | A             |
| Round 3 | D             | B             | F or A        |
| Round 4 | E             | B             | A or F        |

**Case II: F wins 4 prizes.**

In this case everything remains same except B and F will interchange places.

|         | 1st Prize (L) | 2nd Prize (R) | 3rd Prize (C) |
|---------|---------------|---------------|---------------|
| Round 1 | A             | F             | E             |
| Round 2 | F             | C             | A             |
| Round 3 | D             | F             | B or A        |
| Round 4 | E             | F             | A or B        |

Cannot be determined.

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**Directions for questions 32 to 35:** Answer the questions on the basis of the information given below.

Six friends - A, B, C, D, E, and F - participate in an online quiz having 4 rounds. In every round, participants are awarded points and the top three participants of each round get prizes with 1st prize being a LED-TV, 2nd being a Refrigerator and 3rd being a cash prize of Rs. 10,000. All six friends participate in every round of the quiz.

Some details about who-won-what in every round is given below:

- Cash prize is won by only 3 participants and LED-TV is won by a different participant in each round.
- A wins in exactly three rounds and receives two types of prizes. C wins a Refrigerator only.
- There is only one participant who wins a prize in every round but never comes 3rd in any round.
- E wins a prize in two rounds only by coming 1st and 3rd. D wins a LED-TV only.
- D and E never win a prize in the same round. A and C win in the same round.

**Q.35 [11594329]**

Which pair among the following wins at least one same prize?

1 ☐ B and F

2 ☐ A and C

3 ☐ C and E

4 ☐ A and F

**Solution:**

**Correct Answer : 4**

 Answer key/Solution

It is very clear that a total of 12 prizes are distributed in 4 rounds, with 4 each of LED-TV (L), Refrigerator(R) and cash prize of Rs. 10,000(C). Now 'A' has won 2 types of prizes in 3 rounds, this means 'A' has won one of the prizes twice. C and D have won a single prize that is a refrigerator and a LED-TV respectively. Also, E has won two prizes - LED-TV and Cash Prize. This leaves us with two possibilities that either B or F has won 4 prizes.

**Case I:** Let B be the one who won 4 prizes; therefore, F will win one prize only.

We know that LED-TV is won by a different participant in each round, therefore, B must win one LED-TV and 3 – refrigerators. And Now, A won 2 types of prizes, so they must be 1- LED-TV and 2 – cash prizes.

| Friend | Prize      |
|--------|------------|
| A (3)  | L, C, C    |
| B (4)  | L, R, R, R |
| C (1)  | R          |
| D (1)  | L          |
| E (2)  | L, C       |
| F (1)  | C          |

**Case II:** Let F be the one who won 4 prizes; therefore, B will win one prize only.

We know that LED-TV is won by a different participant in each round, therefore, F must win one LED-TV and 3 – refrigerators. And Now, A won 2 types of prizes, so they must be 1- LED-TV and 2 – cash prizes.

| Friend | Prize   |
|--------|---------|
| A (3)  | L, C, C |
| B (1)  | C       |

|       |            |
|-------|------------|
| C (1) | R          |
| D (1) | L          |
| E (2) | L, C       |
| F (4) | L, R, R, R |

**Note:** In above mentioned cases the prizes are won in any order of rounds.

We can make cases for Rounds as well-

**Case I:** B wins 4 prizes.

Let us assume that A, B, D and E won LED-TV in round 1, 2, 3 and 4 respectively. Now since B won first prize in round 2, therefore, second prize must be won in round 1, 3 and 4. And therefore, C won refrigerator in round 2 and since A win in the same round as C, therefore, C won cash prize in round 2. And because D and E never wins a prize in the same round therefore, E must win in round 1.

|         | 1st Prize (L) | 2nd Prize (R) | 3rd Prize (C) |
|---------|---------------|---------------|---------------|
| Round 1 | A             | B             | E             |
| Round 2 | B             | C             | A             |
| Round 3 | D             | B             | F or A        |
| Round 4 | E             | B             | A or F        |

**Case II:** F wins 4 prizes.

In this case everything remains same except B and F will interchange places.

|         | 1st Prize (L) | 2nd Prize (R) | 3rd Prize (C) |
|---------|---------------|---------------|---------------|
| Round 1 | A             | F             | E             |
| Round 2 | F             | C             | A             |
| Round 3 | D             | F             | B or A        |
| Round 4 | E             | F             | A or B        |

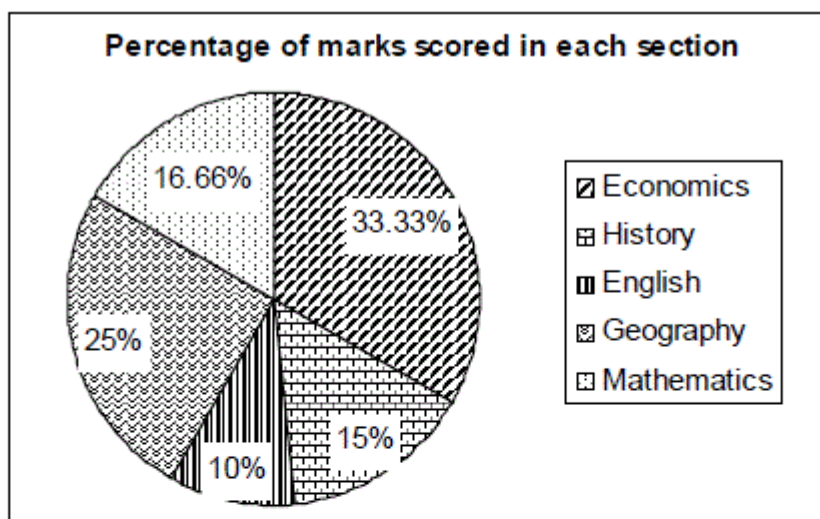
A and F won at least one same prize.

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**Directions for questions 36 to 38:** Answer the questions on the basis of the information given below.

Amy has given an entrance exam which comprises 5 sections: English, Mathematics, Geography, History, and Economics. The cumulative score of these sections will make the final score. In each of the section, the questions asked are of 6 marks, 4 marks, 3 marks, 2 marks and 1 mark. There is no negative marking. The total marks scored by Amy in this exam is less than 350.

The following pie chart gives us the marks scored by Amy in these 5 sections as a percentage of her total marks scored.



The following table provides us the information regarding the percentage of marks scored through questions of 6 marks, 4 marks and 2 marks in the given sections.

|             | 6 marks | 4 marks | 2 marks |
|-------------|---------|---------|---------|
| Mathematics | 36      | 24      | 20      |
| English     | 20      | 0       | 40      |
| Economics   | 12      | 20      | 16      |
| History     | 26.66.. | 26.66.. | 13.33.. |
| Geography   | 40      | 32      | 8       |

### Q.36 [11594329]

What is the total score of Amy in the exam?



**Solution:**

**Correct Answer : 300**

**Your Answer : 300**

We can start this question from any section.

Let us start with Mathematics.

% marks through 6 marks questions = 36%

% marks through 4 marks questions = 24%

% marks through 2 marks questions = 20%

[Answer key/Solution](#)

Taking Ratio  $\rightarrow 36 : 24 : 20$   
 $9 : 6 : 5$

This ratio will represent the ratio of marks scored through 6 marks, 4 marks and 2 marks questions, respectively.

Smallest integers satisfying this ratio will be 9, 6 and 5.

But 9 marks can never be scored through 6 marks questions.

So, we take next multiples i.e., 18, 12, 10. This satisfy that in Mathematics section, 3 questions of 6 marks, 3 questions of 4 marks and 5 questions of 2 marks are done correct.

So, total marks = 40

Now, it is important to understand that this will be the minimum any can score in Mathematics section through 6 marks, 4 marks and 2 marks questions.

Other possibilities will be multiple of 40, we start by smallest value i.e., 40.

80% of marks in Mathematics = 40

So, marks in Mathematics = 50

Now this is 16.66...% of total marks scored by Amy.

$$\frac{1}{6} \times \text{Total marks of Amy} = 50$$

Total marks scored by Amy = 300

[If we take the next multiple of 40, i.e., 80, then total marks scored by Amy becomes 600 which is not possible]

- Marks scored by Amy in Geography = 25% of 300 = 75

$$\text{Marks scored in Geography through 4 marks questions} = \frac{32}{100} \times 75 = 24$$

So, there are 6 questions of 4 marks, which are done correct.

- Similarly, we can find out marks in other subjects.

Marks in Economics = 100

Marks in History = 45

Marks in English = 30

Marks in Geography = 75

Marks in Mathematics = 50

So, marks through 2 mark questions.

$\Rightarrow$  In Economics = 16% of 100 = 16. Therefore, number of questions = 8

$$\Rightarrow \text{In History} = 13.33\% \text{ of } 45 = \frac{2}{15} \times 45 = 6$$

Number of questions = 3

$$\Rightarrow \text{In English} = 40\% \text{ of } 30 = 12$$

Number of questions = 6

$$\Rightarrow \text{In Geography} = 8\% \text{ of } 75 = 8 \times \frac{3}{4} = 6$$

Number of questions = 3

$$\Rightarrow \text{In Mathematics} = \frac{20}{100} \times 50 = 10$$

Number of questions = 5

Total number of 2 marks questions = 8 + 3 + 6 + 3 + 5 = 25

- Marks scored by Amy through 3 marks and 1 mark question  $\rightarrow$

$$\text{In Mathematics} = \frac{20}{100} \times 50 = 10$$

$$\text{In English} = \frac{40}{100} \times 30 = 12$$

$$\text{In Economics} = \frac{52}{100} \times 100 = 52$$

$$\text{In History} = 33.33\% \text{ of } 45 = 15$$

$$\text{In Geography} = \frac{20}{100} \times 75 = 15$$

$$\text{Total marks} = 10 + 12 + 52 + 15 + 15 = 104$$

$$\% \text{ marks} = \frac{104}{300} \times 100 = 34.66\ldots\%$$

The total score of Amy in the exam = 300

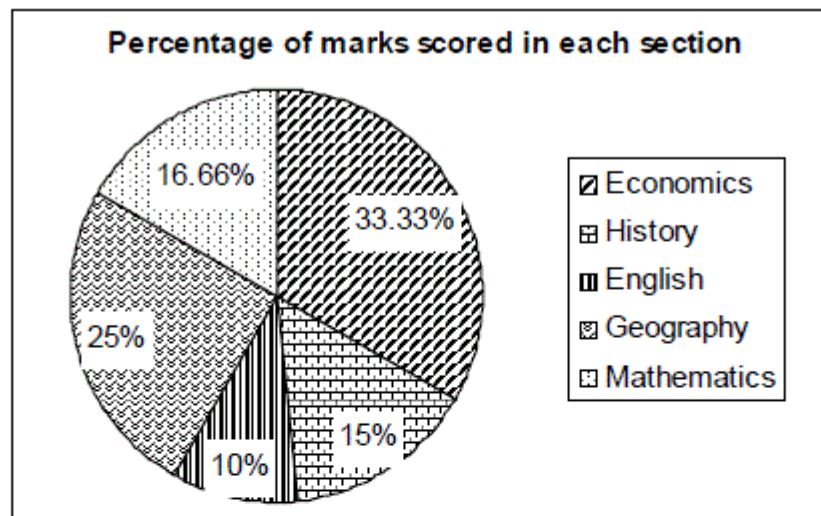
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FeedBack

**Directions for questions 36 to 38:** Answer the questions on the basis of the information given below.

Amy has given an entrance exam which comprises 5 sections: English, Mathematics, Geography, History, and Economics. The cumulative score of these sections will make the final score. In each of the section, the questions asked are of 6 marks, 4 marks, 3 marks, 2 marks and 1 mark. There is no negative marking. The total marks scored by Amy in this exam is less than 350.

The following pie chart gives us the marks scored by Amy in these 5 sections as a percentage of her total marks scored.



The following table provides us the information regarding the percentage of marks scored through questions of 6 marks, 4 marks and 2 marks in the given sections.

|             | 6 marks | 4 marks | 2 marks |
|-------------|---------|---------|---------|
| Mathematics | 36      | 24      | 20      |
| English     | 20      | 0       | 40      |
| Economics   | 12      | 20      | 16      |
| History     | 26.66.. | 26.66.. | 13.33.. |
| Geography   | 40      | 32      | 8       |

**Q.37 [11594329]**

What is the number of 4 marks questions done correctly by Amy in Geography?



**Solution:**

[Answer key/Solution](#)

**Correct Answer : 6****Your Answer : 6**

We can start this question from any section.

Let us start with Mathematics.

% marks through 6 marks questions = 36%

% marks through 4 marks questions = 24%

% marks through 2 marks questions = 20%

Taking Ratio  $\rightarrow 36 : 24 : 20$

$9 : 6 : 5$

This ratio will represent the ratio of marks scored through 6 marks, 4 marks and 2 marks questions, respectively.

Smallest integers satisfying this ratio will be 9, 6 and 5.

But 9 marks can never be scored through 6 marks questions.

So, we take next multiples i.e., 18, 12, 10. This satisfy that in Mathematics section, 3 questions of 6 marks, 3 questions of 4 marks and 5 questions of 2 marks are done correct.

So, total marks = 40

Now, it is important to understand that this will be the minimum any can score in Mathematics section through 6 marks, 4 marks and 2 marks questions.

Other possibilities will be multiple of 40, we start by smallest value i.e., 40.

80% of marks in Mathematics = 40

So, marks in Mathematics = 50

Now this is 16.66...% of total marks scored by Amy.

$$\frac{1}{6} \times \text{Total marks of Amy} = 50$$

Total marks scored by Amy = 300

[If we take the next multiple of 40, i.e., 80, then total marks scored by Amy becomes 600 which is not possible]

- Marks scored by Amy in Geography = 25% of 300 = 75

$$\text{Marks scored in Geography through 4 marks questions} = \frac{32}{100} \times 75 = 24$$

So, there are 6 questions of 4 marks, which are done correct.

- Similarly, we can find out marks in other subjects.

Marks in Economics = 100

Marks in History = 45

Marks in English = 30

Marks in Geography = 75

Marks in Mathematics = 50

So, marks through 2 mark questions.

$\Rightarrow$  In Economics = 16% of 100 = 16. Therefore, number of questions = 8

$\Rightarrow$  In History = 13.33 % of 45 =  $\frac{2}{15} \times 45 = 6$

Number of questions = 3

$\Rightarrow$  In English = 40% of 30 = 12  
Number of questions = 6

$\Rightarrow$  In Geography = 8% of 75 =  $8 \times \frac{3}{4} = 6$   
Number of questions = 3

$\Rightarrow$  In Mathematics =  $\frac{20}{100} \times 50 = 10$

Number of questions = 5

Total number of 2 marks questions = 8 + 3 + 6 + 3 + 5 = 25

- Marks scored by Amy through 3 marks and 1 mark question  $\rightarrow$

$$\text{In Mathematics} = \frac{20}{100} \times 50 = 10$$

$$\text{In English} = \frac{40}{100} \times 30 = 12$$

$$\text{In Economics} = \frac{52}{100} \times 100 = 52$$



100

In History = 33.33% of 45 = 15

$$\text{In Geography} = \frac{20}{100} \times 75 = 15$$

Total marks = 10 + 12 + 52 + 15 + 15 = 104

$$\% \text{ marks} = \frac{104}{300} \times 100 = 34.66\ldots\%$$

The number of 4 marks questions done correctly by Amy in Geography is 6.

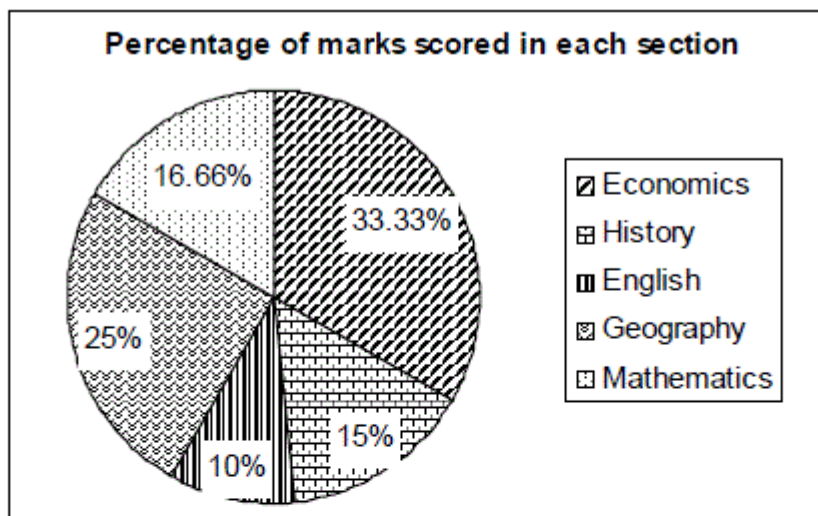
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FeedBack

**Directions for questions 36 to 38:** Answer the questions on the basis of the information given below.

Amy has given an entrance exam which comprises 5 sections: English, Mathematics, Geography, History, and Economics. The cumulative score of these sections will make the final score. In each of the section, the questions asked are of 6 marks, 4 marks, 3 marks, 2 marks and 1 mark. There is no negative marking. The total marks scored by Amy in this exam is less than 350.

The following pie chart gives us the marks scored by Amy in these 5 sections as a percentage of her total marks scored.



The following table provides us the information regarding the percentage of marks scored through questions of 6 marks, 4 marks and 2 marks in the given sections.

|                    | 6 marks | 4 marks | 2 marks |
|--------------------|---------|---------|---------|
| <b>Mathematics</b> | 36      | 24      | 20      |
| <b>English</b>     | 20      | 0       | 40      |
| <b>Economics</b>   | 12      | 20      | 16      |
| <b>History</b>     | 26.66.. | 26.66.. | 13.33.. |
| <b>Geography</b>   | 40      | 32      | 8       |

**Q.38 [11594329]**

The marks scored by Amy through 3 marks and 1 mark questions is what percent of her total score?

1 ☐ 25%

2 ☐ 34.66%

3 ☐ 33.33%

4 ☐ 42%

**Solution:**

**Correct Answer : 2**

**Your Answer : 2**

Answer key/Solution

We can start this question from any section.

Let us start with Mathematics.

% marks through 6 marks questions = 36%

% marks through 4 marks questions = 24%

% marks through 2 marks questions = 20%

Taking Ratio  $\rightarrow 36 : 24 : 20$

$9 : 6 : 5$

This ratio will represent the ratio of marks scored through 6 marks, 4 marks and 2 marks questions, respectively.

Smallest integers satisfying this ratio will be 9, 6 and 5.

But 9 marks can never be scored through 6 marks questions.

So, we take next multiplies i.e., 18, 12, 10. This satisfy that in Mathematics section, 3 questions of 6 marks, 3 questions of 4 marks and 5 questions of 2 marks are done correct.

So, total marks = 40

Now, it is important to understand that this will be the minimum any can score in Mathematics section through 6 marks, 4 marks and 2 marks questions.

Other possibilities will be multiple of 40, we start by smallest value i.e., 40.

80% of marks in Mathematics = 40

So, marks in Mathematics = 50

Now this is 16.66...% of total marks scored by Amy.

$$\frac{1}{6} \times \text{Total marks of Amy} = 50$$

Total marks scored by Amy = 300

[If we take the next multiple of 40, i.e., 80, then total marks scored by Amy becomes 600 which is not possible]

- Marks scored by Amy in Geography = 25% of 300 = 75

$$\text{Marks scored in Geography through 4 marks questions} = \frac{32}{100} \times 75 = 24$$

So, there are 6 questions of 4 marks, which are done correct.

- Similarly, we can find out marks in other subjects.

Marks in Economics = 100

Marks in History = 45

Marks in English = 30

Marks in Geography = 75

Marks in Mathematics = 50

So, marks through 2 mark questions.

$\Rightarrow$  In Economics = 16% of 100 = 16. Therefore, number of questions = 8

$\Rightarrow$  In History = 13.33 % of 45 =  $\frac{2}{15} \times 45 = 6$

Number of questions = 3

$\Rightarrow$  In English = 40% of 30 = 12



Number of questions = 6

$$\Rightarrow \text{In Geography} = 8\% \text{ of } 75 = 8 \times \frac{3}{4} = 6$$

Number of questions = 3

$$\Rightarrow \text{In Mathematics} = \frac{20}{100} \times 50 = 10$$

Number of questions = 5

Total number of 2 marks questions =  $8 + 3 + 6 + 3 + 5 = 25$

- Marks scored by Amy through 3 marks and 1 mark question →

$$\text{In Mathematics} = \frac{20}{100} \times 50 = 10$$

$$\text{In English} = \frac{40}{100} \times 30 = 12$$

$$\text{In Economics} = \frac{52}{100} \times 100 = 52$$

$$\text{In History} = 33.33\% \text{ of } 45 = 15$$

$$\text{In Geography} = \frac{20}{100} \times 75 = 15$$

$$\text{Total marks} = 10 + 12 + 52 + 15 + 15 = 104$$

$$\% \text{ marks} = \frac{104}{300} \times 100 = 34.66\ldots\%$$

The marks scored by Amy through 3 marks and 1 mark questions is 34.66% of her total score.

Bookmark

FeedBack

**Directions for questions 39 to 41:** Answer the questions on the basis of the information given below.

Four professional car racing drivers - Ned, Rob, Sophie and Cathy - participated in a Car Racing Competition at different levels - Beginner, Advanced, Intermediate and Expert (not necessarily in this order). Each of them drove a different car among SX7, i10, Hector and Omni which could further be categorised as Sedan, Hatchback, SUV and Minivan. They finished the competition at different positions - 2nd, 5th, 7th and 8th, in any order.

- If i10 finished at 2nd position, then Cathy finished at 7th position. Rob participated in the Beginner level.
- If Cathy finished at 7th position, then Hector is a SUV otherwise not.
- If Ned competed in the Advanced level, then he finished at 5th position and if he competed in the Intermediate level, he finished at 7th position.
- The car that finished at 8th position was a Hatchback. This was either Hector in which case it competed in the Expert level or Omni in which case it was driven by Cathy.
- If Rob's car was SX7, then SX7 was a Sedan otherwise SX7 was a SUV. Sophie's car was Hector.
- If Hector finished at 7th, then it competed in the Intermediate level otherwise it competed in the Advanced level.
- If SX7 finished at 2nd, then Cathy's car is a Hatchback otherwise Cathy's car is a SUV. Also, if Ned's car is a Minivan, then Ned finished at 7th otherwise Ned finished at 5th.

### Q.39 [11594329]

Who among the following drove i10?

1 ☐ Ned2 ☐ Rob3 ☐ Cathy4 ☐ Cannot be determined**Solution:****Correct Answer : 1** Answer key/Solution

Let us first consider the conditions which are fix and then we will consider the variable conditions.

From statement (i), we know Rob participated in the beginner level. From statement (v), Sophie's car is Hector. From statement (iv), the 8th positioned car is a Hatchback.

From statement (vi), it is clear that Hector competed either at Intermediate level or Advanced level. So, from (iv) we can say Hatchback is at 8th position and it is driven by Cathy.

From statement (vii), we can say that SX7 was at 2nd position. And it is clear that Ned either finished on 5th or 7th position. Consider statement (iii) as well, then we can make 2 cases.

**Case I:**

| Position | Name   | Car    | Car Type  | Level    |
|----------|--------|--------|-----------|----------|
| 2        | Rob    | SX7    | Sedan     | Beginner |
| 5        | Ned    | i10    | SUV       | Advanced |
| 7        | Sophie | Hector | Minivan   |          |
| 8        | Cathy  | Omni   | Hatchback |          |

**Case II:**

| Position | Name   | Car    | Car Type  | Level        |
|----------|--------|--------|-----------|--------------|
| 2        | Rob    | SX7    | Sedan     | Beginner     |
| 5        | Sophie | Hector | SUV       |              |
| 7        | Ned    | i10    | Minivan   | Intermediate |
| 8        | Cathy  | Omni   | Hatchback |              |

From statement (ii), as Cathy has not finished 7th, so Hector is not a SUV.

So, in case 2 Hector is coming out to SUV, thus case 2 is not possible.

From statement (vii), Hector Participated at Intermediate level and thus Cathy at expert level.

Final table looks like→

| Position | Name   | Car    | Car Type  | Level        |
|----------|--------|--------|-----------|--------------|
| 2        | Rob    | SX7    | Sedan     | Beginner     |
| 5        | Ned    | i10    | SUV       | Advanced     |
| 7        | Sophie | Hector | Minivan   | Intermediate |
| 8        | Cathy  | Omni   | Hatchback | Expert       |

i10 was driven by Ned.

Bookmark

FeedBack

**Directions for questions 39 to 41:** Answer the questions on the basis of the information given below.

Four professional car racing drivers - Ned, Rob, Sophie and Cathy - participated in a Car Racing Competition at different levels - Beginner, Advanced, Intermediate and Expert (not necessarily in this order). Each of them drove a different car among SX7, i10, Hector and Omni which could further be categorised as Sedan, Hatchback, SUV and Minivan. They finished the competition at different positions - 2nd, 5th, 7th and 8th, in any order.

- (i) If i10 finished at 2nd position, then Cathy finished at 7th position. Rob participated in the Beginner level.
- (ii) If Cathy finished at 7th position, then Hector is a SUV otherwise not.
- (iii) If Ned competed in the Advanced level, then he finished at 5th position and if he competed in the Intermediate level, he finished at 7th position.
- (iv) The car that finished at 8th position was a Hatchback. This was either Hector in which case it competed in the Expert level or Omni in which case it was driven by Cathy.
- (v) If Rob's car was SX7, then SX7 was a Sedan otherwise SX7 was a SUV. Sophie's car was Hector.
- (vi) If Hector finished at 7th, then it competed in the Intermediate level otherwise it competed in the Advanced level.
- (vii) If SX7 finished at 2nd, then Cathy's car is a Hatchback otherwise Cathy's car is a SUV. Also, if Ned's car is a Minivan, then Ned finished at 7th otherwise Ned finished at 5th.

**Q.40 [11594329]**

Which among the following is a Hatchback?

1 ☐ SX7

2 ☐ i10

3 ☐ Hector

4 ☐ Omni

**Solution:**

**Correct Answer : 4**

 Answer key/Solution

Let us first consider the conditions which are fix and then we will consider the variable conditions.

From statement (i), we know Rob participated in the beginner level. From statement (v), Sophie's car is Hector. From statement (iv), the 8th positioned car is a Hatchback.

From statement (vi), it is clear that Hector competed either at Intermediate level or Advanced level. So, from (iv) we can say Hatchback is at 8th position and it is driven by Cathy.

From statement (vii), we can say that SX7 was at 2nd position. And it is clear that Ned either finished on 5th or 7th position. Consider statement (iii) as well, then we can make 2 cases.

#### Case I:

| Position | Name   | Car    | Car Type  | Level    |
|----------|--------|--------|-----------|----------|
| 2        | Rob    | SX7    | Sedan     | Beginner |
| 5        | Ned    | i10    | SUV       | Advanced |
| 7        | Sophie | Hector | Minivan   |          |
| 8        | Cathy  | Omni   | Hatchback |          |

#### Case II:

| Position | Name   | Car    | Car Type  | Level        |
|----------|--------|--------|-----------|--------------|
| 2        | Rob    | SX7    | Sedan     | Beginner     |
| 5        | Sophie | Hector | SUV       |              |
| 7        | Ned    | i10    | Minivan   | Intermediate |
| 8        | Cathy  | Omni   | Hatchback |              |

From statement (ii), as Cathy has not finished 7th, so Hector is not a SUV.

So, in case 2 Hector is coming out to SUV, thus case 2 is not possible.

From statement (vii), Hector Participated at Intermediate level and thus Cathy at expert level.

Final table looks like→

| Position | Name   | Car    | Car Type  | Level        |
|----------|--------|--------|-----------|--------------|
| 2        | Rob    | SX7    | Sedan     | Beginner     |
| 5        | Ned    | i10    | SUV       | Advanced     |
| 7        | Sophie | Hector | Minivan   | Intermediate |
| 8        | Cathy  | Omni   | Hatchback | Expert       |

Omni is a Hatchback.

[Bookmark](#)
[FeedBack](#)

**Directions for questions 39 to 41:** Answer the questions on the basis of the information given below.

Four professional car racing drivers - Ned, Rob, Sophie and Cathy - participated in a Car Racing Competition at different levels - Beginner, Advanced, Intermediate and Expert (not necessarily in this order). Each of them drove a different car among SX7, i10, Hector and Omni which could further be categorised as Sedan, Hatchback, SUV and Minivan. They finished the competition at different positions - 2nd, 5th, 7th and 8th, in any order.

- (i) If i10 finished at 2nd position, then Cathy finished at 7th position. Rob participated in the Beginner level.
- (ii) If Cathy finished at 7th position, then Hector is a SUV otherwise not.
- (iii) If Ned competed in the Advanced level, then he finished at 5th position and if he competed in the Intermediate level, he finished at 7th position.
- (iv) The car that finished at 8th position was a Hatchback. This was either Hector in which case it competed in the Expert level or Omni in which case it was driven by Cathy.
- (v) If Rob's car was SX7, then SX7 was a Sedan otherwise SX7 was a SUV. Sophie's car was Hector.
- (vi) If Hector finished at 7th, then it competed in the Intermediate level otherwise it competed in the Advanced level.
- (vii) If SX7 finished at 2nd, then Cathy's car is a Hatchback otherwise Cathy's car is a SUV. Also, if Ned's car is a Minivan, then Ned finished at 7th otherwise Ned finished at 5th.

**Q.41 [11594329]**

What is the position at which Cathy finished?

1 ☐ 2nd

2 ☐ 4th

3 ☐ 7th

4 ☐ 8th

**Solution:**

**Correct Answer : 4**

 Answer key/Solution

Let us first consider the conditions which are fix and then we will consider the variable conditions.

From statement (i), we know Rob participated in the beginner level. From statement (v), Sophie's car is Hector. From statement (iv), the 8th positioned car is a Hatchback.

From statement (vi), it is clear that Hector competed either at Intermediate level or Advanced level. So, from (iv) we can say Hatchback is at 8th position and it is driven by Cathy.

From statement (vii), we can say that SX7 was at 2nd position. And it is clear that Ned either finished on 5th or 7th position. Consider statement (iii) as well, then we can make 2 cases.

#### Case I:

| Position | Name   | Car    | Car Type  | Level    |
|----------|--------|--------|-----------|----------|
| 2        | Rob    | SX7    | Sedan     | Beginner |
| 5        | Ned    | i10    | SUV       | Advanced |
| 7        | Sophie | Hector | Minivan   |          |
| 8        | Cathy  | Omni   | Hatchback |          |

#### Case II:

| Position | Name   | Car    | Car Type  | Level        |
|----------|--------|--------|-----------|--------------|
| 2        | Rob    | SX7    | Sedan     | Beginner     |
| 5        | Sophie | Hector | SUV       |              |
| 7        | Ned    | i10    | Minivan   | Intermediate |
| 8        | Cathy  | Omni   | Hatchback |              |

From statement (ii), as Cathy has not finished 7th, so Hector is not a SUV.

So, in case 2 Hector is coming out to SUV, thus case 2 is not possible.

From statement (vii), Hector Participated at Intermediate level and thus Cathy at expert level.

Final table looks like→

| Position | Name   | Car    | Car Type  | Level        |
|----------|--------|--------|-----------|--------------|
| 2        | Rob    | SX7    | Sedan     | Beginner     |
| 5        | Ned    | i10    | SUV       | Advanced     |
| 7        | Sophie | Hector | Minivan   | Intermediate |
| 8        | Cathy  | Omni   | Hatchback | Expert       |

Cathy finished at 8th position.

[Bookmark](#)
[FeedBack](#)

**Directions for questions 42 to 45:** Answer the questions on the basis of the information given below.

There are 5 friends – Jon, Arya, Jaime, Tyrion, and Daenerys. Out of these five, there is only one who eats 4 scoops of icecream everyday, there are at least 2 friends who eat 2 scoops of icecream each everyday and there is at least one who don't eat icecream at all. Further, it is known that:

- (i) The friend who eats 4 scoops of icecream everyday is a Truth teller, who always speaks the truth.
- (ii) Anyone who eats two scoops of icecream is an alternator, who alternates between a truth and a lie.
- (iii) Anyone who does not eat icecream at all is a liar, who always speaks a lie.
- (iv) Anyone who eats icecream, he eats icecream of one of the 3 flavours – Black Current, Chocolate and Vanilla and no one eats icecream of more than one flavour.
- (v) Each of the 3 flavours of icecream is eaten by at least one of the 5 friends.

Further, each of the 5 friends made two statements as below:

**Arya:**

- (1) Everyday, I eat 2 scoops of icecream, one each of Black Current and Chocolate.
- (2) Jaime eats either 2 scoops of Vanilla or 2 scoops of Black Current everyday.

**Jon:**

- (1) I do not eat icecream at all.
- (2) Either Tyrion eats 4 scoops of Black Current everyday or Arya eats 4 scoops of Vanilla everyday.

**Jaime:**

- (1) I eat 4 scoops of Chocolate everyday.
- (2) Daenerys eats 2 scoops of Vanilla everyday.

**Daenerys:**

- (1) I eat 2 scoops of Chocolate everyday.
- (2) Jaime eats 2 scoops of Chocolate everyday.

**Tyrion:**

- (1) Jon eats 2 scoops of Vanilla everyday.
- (2) Arya does not eat icecream at all.

---

**Q.42 [11594329]**

Who among the following is the truth teller?

---

1 ☐ Jon

---

2 ☐ Jaime

---

3 ☐ Tyrion

---

4 ☐ Daenerys

---

**Solution:****Correct Answer : 3** Answer key/Solution

Here the key statement is the statement by Jon – “I do not eat icecream at all”. Anyone who does not eat icecream cannot make this statement as it would then be a true statement whereas it is given that anyone who does not eat icecream, he always lies (Refer point iii). At the same time he cannot be the truth teller also. So, the only possibility left is that he is an alternator, whose first statement cannot be true. This implies that his other statement must be true, which is “Either Tyrion eats 4 scoops of Black Current or Arya eats 4 scoops of Vanilla everyday”. Now, since we know that Arya cannot be a truth teller as she claims to eat 2 different flavours of icecream, which is a violation to point (iv). Therefore, Tyrion must be the truth teller.

(Note: Anyone who eats 4 scoops of icecream is a truth teller (Refer point i) and it is given there is only one such person among the 5 friends). Now, since Tyrion is the truth teller therefore, both his statements must be true. Hence, it can be concluded that Jon eats 2 scoops of icecream everyday and that Arya does not eat icecream, therefore Arya always lies and hence both of her statements should be false.

Till now, the following can be concluded:

Liar → Arya → does not eat icecream.

Alternator → Jon → 2 scoops of Vanilla.

Truth Teller → Tyrion → 4 scoops of Black Current.

Now considering point (v), there must be at least one of the remaining two friends – Jaime and Daenerys, who eats Chocolate icecream and there must be at least one more alternator. On seeing the two statements made by Daenerys one must be true and the other statement must be false, as both statements cannot be false because then there will be no one who eats Chocolate icecream and both statements cannot be true as well since there is only one truth teller.

Case I: Statement (1) made by Daenerys is a truth and statement (2) is a lie.

This implies that 2nd statement of Jaime is a lie. And Jaime's 1st statement cannot be true as well as (because anyone who eats 4 scoops of icecream is a truth teller). Hence, both statements of Jaime are false. Hence, Jaime is a liar.

Case II: Statement (1) made by Daenerys is a lie and statement (2) is a truth.

This implies that Jaime is an alternator as he eats 2 scoops of Chocolate icecream and as the 1st statement of Jaime is a lie hence, 2nd statement of his must be true i.e., Daenerys eats 2 scoops of Vanilla.

Based on all this, 2 cases are possible:

Case I:

Arya → Liar → does not eat icecream.

Jon → Alternator → eats 2 scoops of Vanilla.

Jaime → Liar → does not eat icecream.

Daenerys → Alternator → eats 2 scoops of Chocolate icecream.

Tyrion → Truth Teller → eats 4 scoops of Black Current.

Case II:

Arya → Liar → does not eat icecream.

Jon → Alternator → eats 2 scoops of Vanilla.

Jaime → Alternator → eats 2 scoops of Chocolate icecream.

Daenerys → Alternator → eats 2 scoops of Vanilla.

Tyrion → Truth Teller → eats 4 scoops of Black Current.

Tyrion is the truth teller.

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**Directions for questions 42 to 45:** Answer the questions on the basis of the information given below.

There are 5 friends – Jon, Arya, Jaime, Tyrion, and Daenerys. Out of these five, there is only one who eats 4 scoops of icecream everyday, there are at least 2 friends who eat 2 scoops of icecream each everyday and there is at least one who don't eat icecream at all. Further, it is known that:

- (i) The friend who eats 4 scoops of icecream everyday is a Truth teller, who always speaks the truth.
- (ii) Anyone who eats two scoops of icecream is an alternator, who alternates between a truth and a lie.
- (iii) Anyone who does not eat icecream at all is a liar, who always speaks a lie.
- (iv) Anyone who eats icecream, he eats icecream of one of the 3 flavours – Black Current, Chocolate and Vanilla and no one eats icecream of more than one flavour.
- (v) Each of the 3 flavours of icecream is eaten by at least one of the 5 friends.

Further, each of the 5 friends made two statements as below:

**Arya:**

- (1) Everyday, I eat 2 scoops of icecream, one each of Black Current and Chocolate.
- (2) Jaime eats either 2 scoops of Vanilla or 2 scoops of Black Current everyday.

**Jon:**

- (1) I do not eat icecream at all.
- (2) Either Tyrion eats 4 scoops of Black Current everyday or Arya eats 4 scoops of Vanilla everyday.

**Jaime:**

- (1) I eat 4 scoops of Chocolate everyday.
- (2) Daenerys eats 2 scoops of Vanilla everyday.

**Daenerys:**

- (1) I eat 2 scoops of Chocolate everyday.
- (2) Jaime eats 2 scoops of Chocolate everyday.

**Tyrion:**

- (1) Jon eats 2 scoops of Vanilla everyday.
- (2) Arya does not eat icecream at all.

---

**Q.43 [11594329]**

Which of the following is true about Jon?

- 1 ☐ Does not eat icecream at all.
  - 2 ☐ Eats 2 scoops of Vanilla everyday.
  - 3 ☐ Eats 2 scoops of Chocolate everyday.
  - 4 ☐ Eats 4 scoops of Black Current everyday.
-

**Solution:****Correct Answer : 2** Answer key/Solution

Here the key statement is the statement by Jon – “I do not eat icecream at all”. Anyone who does not eat icecream cannot make this statement as it would then be a true statement whereas it is given that anyone who does not eat icecream, he always lies (Refer point iii). At the same time he cannot be the truth teller also. So, the only possibility left is that he is an alternator, whose first statement cannot be true. This implies that his other statement must be true, which is “Either Tyrion eats 4 scoops of Black Current or Arya eats 4 scoops of Vanilla everyday”. Now, since we know that Arya cannot be a truth teller as she claims to eat 2 different flavours of icecream, which is a violation to point (iv). Therefore, Tyrion must be the truth teller.

(Note: Anyone who eats 4 scoops of icecream is a truth teller (Refer point i) and it is given there is only one such person among the 5 friends). Now, since Tyrion is the truth teller therefore, both his statements must be true. Hence, it can be concluded that Jon eats 2 scoops of icecream everyday and that Arya does not eat icecream, therefore Arya always lies and hence both of her statements should be false.

Till now, the following can be concluded:

Liar → Arya → does not eat icecream.

Alternator → Jon → 2 scoops of Vanilla.

Truth Teller → Tyrion → 4 scoops of Black Current.

Now considering point (v), there must be at least one of the remaining two friends – Jaime and Daenerys, who eats Chocolate icecream and there must be at least one more alternator. On seeing the two statements made by Daenerys one must be true and the other statement must be false, as both statements cannot be false because then there will be no one who eats Chocolate icecream and both statements cannot be true as well since there is only one truth teller.

Case I: Statement (1) made by Daenerys is a truth and statement (2) is a lie.

This implies that 2nd statement of Jaime is a lie. And Jaime's 1st statement cannot be true as well as (because anyone who eats 4 scoops of icecream is a truth teller). Hence, both statements of Jaime are false. Hence, Jaime is a liar.

Case II: Statement (1) made by Daenerys is a lie and statement (2) is a truth.

This implies that Jaime is an alternator as he eats 2 scoops of Chocolate icecream and as the 1st statement of Jaime is a lie hence, 2nd statement of his must be true i.e., Daenerys eats 2 scoops of Vanilla.

Based on all this, 2 cases are possible:

Case I:

Arya → Liar → does not eat icecream.

Jon → Alternator → eats 2 scoops of Vanilla.

Jaime → Liar → does not eat icecream.

Daenerys → Alternator → eats 2 scoops of Chocolate icecream.

Tyrion → Truth Teller → eats 4 scoops of Black Current.

Case II:

Arya → Liar → does not eat icecream.

Jon → Alternator → eats 2 scoops of Vanilla.

Jaime → Alternator → eats 2 scoops of Chocolate icecream.

Daenerys → Alternator → eats 2 scoops of Vanilla.

Tyrion → Truth Teller → eats 4 scoops of Black Current.

Jon eats 2 scoops of Vanilla everyday.

[Bookmark](#)[FeedBack](#)

**Directions for questions 42 to 45:** Answer the questions on the basis of the information given below.

There are 5 friends – Jon, Arya, Jaime, Tyrion, and Daenerys. Out of these five, there is only one who eats 4 scoops of icecream everyday, there are at least 2 friends who eat 2 scoops of icecream each everyday and there is at least one who don't eat icecream at all. Further, it is known that:

- (i) The friend who eats 4 scoops of icecream everyday is a Truth teller, who always speaks the truth.
- (ii) Anyone who eats two scoops of icecream is an alternator, who alternates between a truth and a lie.
- (iii) Anyone who does not eat icecream at all is a liar, who always speaks a lie.
- (iv) Anyone who eats icecream, he eats icecream of one of the 3 flavours – Black Current, Chocolate and Vanilla and no one eats icecream of more than one flavour.
- (v) Each of the 3 flavours of icecream is eaten by at least one of the 5 friends.

Further, each of the 5 friends made two statements as below:

**Arya:**

- (1) Everyday, I eat 2 scoops of icecream, one each of Black Current and Chocolate.
- (2) Jaime eats either 2 scoops of Vanilla or 2 scoops of Black Current everyday.

**Jon:**

- (1) I do not eat icecream at all.
- (2) Either Tyrion eats 4 scoops of Black Current everyday or Arya eats 4 scoops of Vanilla everyday.

**Jaime:**

- (1) I eat 4 scoops of Chocolate everyday.
- (2) Daenerys eats 2 scoops of Vanilla everyday.

**Daenerys:**

- (1) I eat 2 scoops of Chocolate everyday.
- (2) Jaime eats 2 scoops of Chocolate everyday.

**Tyrion:**

- (1) Jon eats 2 scoops of Vanilla everyday.
- (2) Arya does not eat icecream at all.

---

**Q.44 [11594329]**

Who among the following options could be the one who eats Vanilla?

1 ☐ Arya

2 ☐ Daenerys

3 ☐ Tyrion

4 ☐ Jaime

---

**Solution:****Correct Answer : 2** Answer key/Solution

Here the key statement is the statement by Jon – “I do not eat icecream at all”. Anyone who does not eat icecream cannot make this statement as it would then be a true statement whereas it is given that anyone who does not eat icecream, he always lies (Refer point iii). At the same time he cannot be the truth teller also. So, the only possibility left is that he is an alternator, whose first statement cannot be true. This implies that his other statement must be true, which is “Either Tyrion eats 4 scoops of Black Current or Arya eats 4 scoops of Vanilla everyday”. Now, since we know that Arya cannot be a truth teller as she claims to eat 2 different flavours of icecream, which is a violation to point (iv). Therefore, Tyrion must be the truth teller.

(Note: Anyone who eats 4 scoops of icecream is a truth teller (Refer point i) and it is given there is only one such person among the 5 friends). Now, since Tyrion is the truth teller therefore, both his statements must be true. Hence, it can be concluded that Jon eats 2 scoops of icecream everyday and that Arya does not eat icecream, therefore Arya always lies and hence both of her statements should be false.

Till now, the following can be concluded:

Liar → Arya → does not eat icecream.

Alternator → Jon → 2 scoops of Vanilla.

Truth Teller → Tyrion → 4 scoops of Black Current.

Now considering point (v), there must be at least one of the remaining two friends – Jaime and Daenerys, who eats Chocolate icecream and there must be at least one more alternator. On seeing the two statements made by Daenerys one must be true and the other statement must be false, as both statements cannot be false because then there will be no one who eats Chocolate icecream and both statements cannot be true as well since there is only one truth teller.

Case I: Statement (1) made by Daenerys is a truth and statement (2) is a lie.

This implies that 2nd statement of Jaime is a lie. And Jaime's 1st statement cannot be true as well as (because anyone who eats 4 scoops of icecream is a truth teller). Hence, both statements of Jaime are false. Hence, Jaime is a liar.

Case II: Statement (1) made by Daenerys is a lie and statement (2) is a truth.

This implies that Jaime is an alternator as he eats 2 scoops of Chocolate icecream and as the 1st statement of Jaime is a lie hence, 2nd statement of his must be true i.e., Daenerys eats 2 scoops of Vanilla.

Based on all this, 2 cases are possible:

Case I:

Arya → Liar → does not eat icecream.

Jon → Alternator → eats 2 scoops of Vanilla.

Jaime → Liar → does not eat icecream.

Daenerys → Alternator → eats 2 scoops of Chocolate icecream.

Tyrion → Truth Teller → eats 4 scoops of Black Current.

Case II:

Arya → Liar → does not eat icecream.

Jon → Alternator → eats 2 scoops of Vanilla.

Jaime → Alternator → eats 2 scoops of Chocolate icecream.

Daenerys → Alternator → eats 2 scoops of Vanilla.

Tyrion → Truth Teller → eats 4 scoops of Black Current.

Daenerys could be the one who eats Vanilla (Refer case 2).

Bookmark

FeedBack

**Directions for questions 42 to 45:** Answer the questions on the basis of the information given below.

There are 5 friends – Jon, Arya, Jaime, Tyrion, and Daenerys. Out of these five, there is only one who eats 4 scoops of icecream everyday, there are at least 2 friends who eat 2 scoops of icecream each everyday and there is at least one who don't eat icecream at all. Further, it is known that:

- (i) The friend who eats 4 scoops of icecream everyday is a Truth teller, who always speaks the truth.
- (ii) Anyone who eats two scoops of icecream is an alternator, who alternates between a truth and a lie.
- (iii) Anyone who does not eat icecream at all is a liar, who always speaks a lie.
- (iv) Anyone who eats icecream, he eats icecream of one of the 3 flavours – Black Current, Chocolate and Vanilla and no one eats icecream of more than one flavour.
- (v) Each of the 3 flavours of icecream is eaten by at least one of the 5 friends.

Further, each of the 5 friends made two statements as below:

**Arya:**

- (1) Everyday, I eat 2 scoops of icecream, one each of Black Current and Chocolate.
- (2) Jaime eats either 2 scoops of Vanilla or 2 scoops of Black Current everyday.

**Jon:**

- (1) I do not eat icecream at all.
- (2) Either Tyrion eats 4 scoops of Black Current everyday or Arya eats 4 scoops of Vanilla everyday.

**Jaime:**

- (1) I eat 4 scoops of Chocolate everyday.
- (2) Daenerys eats 2 scoops of Vanilla everyday.

**Daenerys:**

- (1) I eat 2 scoops of Chocolate everyday.
- (2) Jaime eats 2 scoops of Chocolate everyday.

**Tyrion:**

- (1) Jon eats 2 scoops of Vanilla everyday.
- (2) Arya does not eat icecream at all.

---

**Q.45 [11594329]**

Jaime is a/an

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1 ☐ Liar

---

2 ☐ Alternator

---

3 ☐ Liar or alternator

---

4 ☐ Liar or truth teller

---



**Solution:****Correct Answer : 3** Answer key/Solution

Here the key statement is the statement by Jon – “I do not eat icecream at all”. Anyone who does not eat icecream cannot make this statement as it would then be a true statement whereas it is given that anyone who does not eat icecream, he always lies (Refer point iii). At the same time he cannot be the truth teller also. So, the only possibility left is that he is an alternator, whose first statement cannot be true. This implies that his other statement must be true, which is “Either Tyrion eats 4 scoops of Black Current or Arya eats 4 scoops of Vanilla everyday”. Now, since we know that Arya cannot be a truth teller as she claims to eat 2 different flavours of icecream, which is a violation to point (iv). Therefore, Tyrion must be the truth teller.

(Note: Anyone who eats 4 scoops of icecream is a truth teller (Refer point i) and it is given there is only one such person among the 5 friends). Now, since Tyrion is the truth teller therefore, both his statements must be true. Hence, it can be concluded that Jon eats 2 scoops of icecream everyday and that Arya does not eat icecream, therefore Arya always lies and hence both of her statements should be false.

Till now, the following can be concluded:

Liar  $\rightarrow$  Arya  $\rightarrow$  does not eat icecream.

Alternator  $\rightarrow$  Jon  $\rightarrow$  2 scoops of Vanilla.

Truth Teller  $\rightarrow$  Tyrion  $\rightarrow$  4 scoops of Black Current.

Now considering point (v), there must be at least one of the remaining two friends – Jaime and Daenerys, who eats Chocolate icecream and there must be at least one more alternator. On seeing the two statements made by Daenerys one must be true and the other statement must be false, as both statements cannot be false because then there will be no one who eats Chocolate icecream and both statements cannot be true as well since there is only one truth teller.

Case I: Statement (1) made by Daenerys is a truth and statement (2) is a lie.

This implies that 2nd statement of Jaime is a lie. And Jaime's 1st statement cannot be true as well as (because anyone who eats 4 scoops of icecream is a truth teller). Hence, both statements of Jaime are false. Hence, Jaime is a liar.

Case II: Statement (1) made by Daenerys is a lie and statement (2) is a truth.

This implies that Jaime is an alternator as he eats 2 scoops of Chocolate icecream and as the 1st statement of Jaime is a lie hence, 2nd statement of his must be true i.e., Daenerys eats 2 scoops of Vanilla.

Based on all this, 2 cases are possible:

Case I:

Arya  $\rightarrow$  Liar  $\rightarrow$  does not eat icecream.

Jon  $\rightarrow$  Alternator  $\rightarrow$  eats 2 scoops of Vanilla.

Jaime  $\rightarrow$  Liar  $\rightarrow$  does not eat icecream.

Daenerys  $\rightarrow$  Alternator  $\rightarrow$  eats 2 scoops of Chocolate icecream.

Tyrion  $\rightarrow$  Truth Teller  $\rightarrow$  eats 4 scoops of Black Current.

Case II:

Arya  $\rightarrow$  Liar  $\rightarrow$  does not eat icecream.

Jon  $\rightarrow$  Alternator  $\rightarrow$  eats 2 scoops of Vanilla.

Jaime  $\rightarrow$  Alternator  $\rightarrow$  eats 2 scoops of Chocolate icecream.

Daenerys  $\rightarrow$  Alternator  $\rightarrow$  eats 2 scoops of Vanilla.

Tyrion  $\rightarrow$  Truth Teller  $\rightarrow$  eats 4 scoops of Black Current.

Jaime is either a liar or an alternator.

Bookmark

FeedBack

**Sec 3****Q.46 [11594329]**

Amit, Shashi, Niki, and Nishant have some candies with each of them. Two times the number of candies with Amit equals three times the number of candies with Niki, while two times the number of candies with Niki equals three times the number of candies with Shashi and 5 times the number of candies with Shashi equals 4 times the number of candies with Nishant. What is the minimum number of candies that can be there with all 4 of them put together?

---

1 ☐ 19

---

2 ☐ 131

---

3 ☐ 24

---

4 ☐ None of these



**Solution:**

**Correct Answer : 3**

**Your Answer : 3**

Answer key/Solution

Let the number of candies with Amit, Shashi, Niki and Nishant be  $a$ ,  $b$ ,  $c$  and  $d$  respectively.

According to the given relation, we get

$$\Rightarrow 2a = 3c \quad \rightarrow (1) \times 2 \rightarrow 4a = 6c$$

$$\Rightarrow 2c = 3b \quad \rightarrow (2) \times 3 \rightarrow 6c = 9b$$

$$\Rightarrow 5b = 4d \quad \rightarrow (3)$$

From (1) and (2), we get  $4a = 6c = 9b$  ... (4)

Multiplying eq.(3)  $\times 9$  and (4)  $\times 5$ , we get

$$45b = 36d$$

$$\text{and } 20a = 30c = 45b$$

Combining the two,

$$\Rightarrow 20a = 30c = 45b = 36d$$

$$\Rightarrow a : b : c : d = 9 : 4 : 6 : 5$$

Hence, minimum number of candies with all of them put together =  $9 + 4 + 6 + 5 = 24$ .

Bookmark

FeedBack

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#### Q.47 [11594329]

Two riders in a circus around a circular track meet each other every 54 minutes while driving in the same direction and every 9 minutes when driving in opposite directions. Had the length of the circular track been reduced to 400 m, they would have met 1 minute sooner than usual when driving in opposite directions. Find the speed of the slower rider.

---

1 ☐ 24.33 m/min

---

2 ☐ 22 m/min

---

3 ☐ 26.46 m/min

---

4 ☐ 20.83 m/min

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**Solution:****Correct Answer : 4**[Answer key/Solution](#)

The sum of speed of two riders when riding in opposite direction is  $= \frac{400 \text{ m}}{8 \text{ minutes}} = 50 \text{ m/min}$

Length of the original circular track =  $9 \times 50 = 450 \text{ m}$

Difference of speed =  $\frac{450 \text{ m}}{54 \text{ minutes}} = 8.33 \text{ m/min}$

We can calculate to get the speed of the slower driver to be 20.83 m/min.

Bookmark

FeedBack

**Q.48 [11594329]**

If the value of the expression  $\frac{9^{25-a} 15^{11+b} 2^{13-a}}{30^{10-a+b}}$  is an integer, then find the maximum value of  $(a + b)$ .

**Solution:****Correct Answer : 54**[Answer key/Solution](#)

$$\frac{9^{25-a} 15^{11+b} 2^{13-a}}{30^{10-a+b}} = \frac{3^{50-a+11+b} 5^{11+b} 2^{13-a}}{3^{10-a+b} 5^{10-a+b} 2^{10-a+b}} = 3^{51-a} 2^{3-b} 5^{1+a}$$

For the maximum value of  $(a + b)$  such that the given expression is an integer will be, when  $a = 51$  and  $b = 3$ .  
Therefore, answer is 54.

Bookmark

FeedBack

**Q.49 [11594329]**

In a year, a shopkeeper imported the number of suitcases of three brands- VIP, Safari and Samsonite in the ratio of 8 : 6 : 11. In the next year, the number of suitcases imported of VIP and Samsonite got increased by 37.5% and 18.18%, respectively and that of Safari got decreased by 16.66%. If the difference between the number of imported suitcases of Samsonite and Safari was 5120 in the next year, then find the total number of suitcases imported in the next year?





**Solution:****Correct Answer : 18560****Your Answer : 18560**[Answer key/Solution](#)

Let the number of Suitcases imported of VIP, Safari and Samsonite be  $8x$ ,  $6x$  and  $11x$ , respectively.

In the next year, the number of suitcases imported of VIP was  $= \left(1 + \frac{3}{8}\right) \times 8x = \left(\frac{11}{8}\right) \times 8x = 11x$

And that of Samsonite  $= \left(1 + \frac{2}{11}\right) \times 11x = \left(\frac{13}{11}\right) \times 11x = 13x$

And that of Safari  $= \left(1 - \frac{1}{6}\right) \times 6x = \left(\frac{5}{6}\right) \times 6x = 5x$

According to the question,

$$13x - 5x = 5120$$

$$\Rightarrow x = 640$$

Hence, the total number of suitcases imported in the next year was  $= (11x + 13x + 5x) = 29x = 18560$ .

Bookmark

FeedBack

**Q.50 [11594329]**

The average marks of  $\frac{3}{7}$ th of the entire class is 70% of the average marks of the entire class. What is the ratio of the average marks of the remaining students of class to that of the entire class?

1 ☐ 49 : 402 ☐ 3 : 73 ☐ 7 : 44 ☐ None of these**Solution:****Correct Answer : 1****Your Answer : 1**[Answer key/Solution](#)

Let the number of students = 7 and the average marks of the class = 100;  
Sum of marks =  $100 \times 7 = 700$

So,  $\frac{3}{7}$ th of entire class = 3 students

It is given that, average of  $\frac{3}{7}$ th of class is 70% of average marks of the class

$$\Rightarrow \text{Sum of their marks (i.e., 3 students)} = 3 \times 70 = 210$$

$$\text{Sum of marks of remaining class (i.e., remaining 4 students)} = 700 - 210 = 490$$

$$\text{Ratio of average marks of remaining class to that of entire class} = \frac{490}{4} : 100 = 49 : 40$$

Bookmark

FeedBack

**Q.51 [11594329]**

If the LCM of  $\{1, 2, 3, 4, \dots, 81\} = P$ , then by what number should  $P$  be multiplied so that the resultant becomes the LCM of  $\{1, 2, 3, 4, \dots, 87\}$ ?

1 ☐ 1662 ☐ 2493 ☐ 834 ☐ 87**Solution:****Correct Answer : 3**

LCM of  $\{1, 2, 3, \dots, 81\} = P$   
 Let us include 82.  
 $82 = 2 \times 41$   
 We can say both 2 and 41 are included in  $P$ .  
 So LCM of  $\{1, 2, 3, \dots, 81, 82\} = P$   
 Now 83 is prime  
 So LCM of  $\{1, 2, 3, 82, 83\} = P \times 83$   
 Now  $84 = 2^2 \times 3 \times 7$  all included in  $P$ .  
 $85 = 5 \times 17$ , both included in  $P$ .  
 $86 = 2 \times 43$ , both included in  $P$ .  
 $87 = 3 \times 29$ , both included in  $P$ .  
 So LCM of  $\{1, 2, 3, \dots, 87\} = 83 \times P$ .

Bookmark

FeedBack

 Answer key/Solution**Q.52 [11594329]**

A shopkeeper purchases basmati rice from a wholesaler. In his storage, rats eat away 10% of the rice. He mixes impurities to the extent of 20% of the rice present and gave a price mark up of 25%. Further while selling he sells at a discount of 16.67% and his balance reads 1200 grams for 1kg. Find his profit/loss percentage.

**Solution:****Correct Answer : 35** Answer key/Solution

Since multiple stages of percentage change are involved in this question so we will solve it using multiplying factors corresponding to various percentage changes. So effective multiplying factor here =  $\frac{9}{10} \times \frac{6}{5} \times \frac{5}{4} \times \frac{5}{6} \times \frac{12}{10} = \frac{27}{20} = 1 + \frac{7}{20}$

Hence percentage increase =  $\frac{7}{20} \times 100 = 35\%$

So, he made a profit of 35%.

Bookmark

FeedBack

**Q.53 [11594329]**

The sum of the series  $\log(8/25) + \log(16/125) + \dots$  upto 8 terms is

1 ☐  $4(11\log 2 - 13\log 5)$

2 ☐  $4(13\log 2 - 11\log 5)$

3 ☐  $3(11\log 2 - 17\log 5)$

4 ☐  $5(7\log 2 - 13\log 5)$

**Solution:**

**Correct Answer : 2**

$$\begin{aligned} \log\left(\frac{8}{25}\right) + \log\left(\frac{16}{125}\right) + \dots &= \log \frac{2^3}{5^2} + \log \frac{2^4}{5^3} + \dots \\ &= 3\log 2 - 2\log 5 + 4\log 2 - 3\log 5 + \dots \\ &= (3\log 2 + 4\log 2 + \dots \text{ upto 8 terms }) - (2\log 5 + 3\log 5 + \dots \text{ upto 8 terms }) \\ &= 52\log 2 - 44\log 5 = 4(13\log 2 - 11\log 5). \end{aligned}$$

Bookmark

FeedBack

 Answer key/Solution

**Q.54 [11594329]**

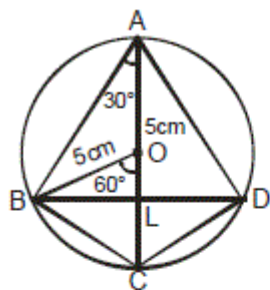
A, B, C, and D are points on the circumference of a circle of center 'O' and radius 5 cm, such that ABD is an equilateral triangle and AC is a diameter of the circle. What is the ratio of area of  $\triangle ABD$  to that of  $\triangle BCD$ ?

1 ☐ 2 : 1

2 ☐ 3 : 1

3 ☐ 4 : 1

4 ☐ 3 : 2

**Solution:****Correct Answer : 2**[Answer key/Solution](#)

$$\angle BAO = 30^\circ$$

$$\angle BOC = 60^\circ \text{ (Angle at the center)}$$

$$\angle OBC = \angle OCB = 60^\circ$$

Therefore,  $BC = 5 \text{ cm}$

Using sine rule in  $\triangle AOB$ ,

$$\frac{5}{\sin 30^\circ} = \frac{AB}{\sin 120^\circ}$$

$$\Rightarrow \frac{5}{\frac{1}{2}} = \frac{AB}{\frac{\sqrt{3}}{2}}$$

$$\Rightarrow AB = 5\sqrt{3} \text{ cm}$$

$$\text{Area of } \triangle ABD = \frac{\sqrt{3}}{4} \times (5\sqrt{3})^2 = \frac{\sqrt{3}}{4} \times 75 \text{ cm}^2$$

$$\text{Area of } \triangle BCD = 2 \times \text{Area of } \triangle BLC = 2 \times \frac{1}{2} \times \text{Area of } \triangle BOC = \frac{\sqrt{3}}{4} \times 5^2 \text{ cm}^2$$

$$\text{Hence, required ratio} = \frac{\sqrt{3}}{4} \times 75 : \frac{\sqrt{3}}{4} \times 5^2 = 3 : 1.$$

Bookmark

FeedBack

**Q.55 [11594329]**

Three pipes P1, P2, and P3 can fill a tank in 15, 20 and 24 minutes, respectively. If P2 is open all the time and P1 and P3 are open for one minute each alternatively, then the five-sixth of the tank will be full in how many minutes?

**Solution:****Correct Answer : 8**[Answer key/Solution](#)

P1 and P2 can fill the tank in one minute =  $\frac{1}{15} + \frac{1}{20} = \frac{7}{60}$  part

P2 and P3 can fill the tank in one minute =  $\frac{1}{20} + \frac{1}{24} = \frac{11}{120}$  part

Part of tank filled in two minutes =  $\frac{7}{60} + \frac{11}{120} = \frac{25}{120}$

Part of the tank filled in 8 minutes =  $\frac{25}{120} \times 4 = \frac{100}{120} = \frac{5}{6}$ .

Hence, the answer is 8 minutes.

Bookmark

FeedBack

**Q.56 [11594329]**

The area of a rectangle is  $899 \text{ cm}^2$  where the length and breadth of the rectangle are integral multiple of cm. If the rectangle is changed to a square either by increasing its breadth only or by decreasing its length only (but not both), which of the following could not be the change in its area?

1 ☐  $58 \text{ cm}^2$ 2 ☐  $62 \text{ cm}^2$ 3 ☐  $900 \text{ cm}^2$ 4 ☐  $807302 \text{ cm}^2$ **Solution:****Correct Answer : 3**[Answer key/Solution](#)

$899 = 899 \times 1$  or  $31 \times 29$

Case I: Length = 899 and breadth = 1

(a) Making it a square by increasing breadth to 899  $\Rightarrow L = B = 899$

Hence, area =  $899 \times 899 = 808201$

Therefore, an increase of = 807302

(b) Making it a square by decreasing the length to 1  $\Rightarrow L = B = 1$

Area =  $1 \times 1 = 1$

Therefore, a decrease of 898 i.e.  $(899 - 1)$

Case II: Length = 31 and breadth = 29

(a) Increasing the breadth to 31  $\Rightarrow L = B = 31$

Area =  $31 \times 31 = 961$

Hence, an increase of 62

(b) Decreasing the length to 29 to make it a square  $\Rightarrow L = B = 29$

Area =  $29 \times 29 = 841$

Hence, a decrease of 58.

Bookmark

FeedBack

**Q.57 [11594329]**

For a positive integer  $x$ ,  $f(x + 2) = x/2 + f(x)$ , when  $x$  is even and  $f(x + 2) = x + f(x)$ , when  $x$  is odd. If  $f(1) = 2$  and  $f(2) = 5$ , then find the value of  $f(f(f(5))) + f(f(f(6)))$ .

**Solution:****Correct Answer : 38** Answer key/Solution

For an even positive integer  $x$ ,  $f(x + 2) = \frac{x}{2} + f(x)$

For an odd positive integer  $x$ ,  $f(x + 2) = x + f(x)$

Let's find  $f(f(f(5)))$ .

$$f(5) = 3 + f(3) = 3 + [f(1) + 1] = 3 + 3 = 6$$

$$f(f(5)) = f(6) = \frac{4}{2} + f(4) = 2 + [\frac{2}{2} + f(2)] = 8$$

$$f(f(f(5))) = f(8) = \frac{6}{2} + f(6) = 3 + 8 = 11$$

Similarly, we can find  $f(f(f(6)))$ ,  $f(6) = 8$ , so  $f(f(6)) = f(8) = 11$  (We have found these values in the previous steps)

$$f(f(f(6))) = f(11) = 9 + f(9) = 9 + [7 + f(7)] = 9 + [7 + \{5 + f(5)\}]$$

Since  $f(5) = 6$  (Calculated earlier)

$$f(f(f(6))) = 27$$

$$\text{Required sum} = 11 + 27 = 38.$$

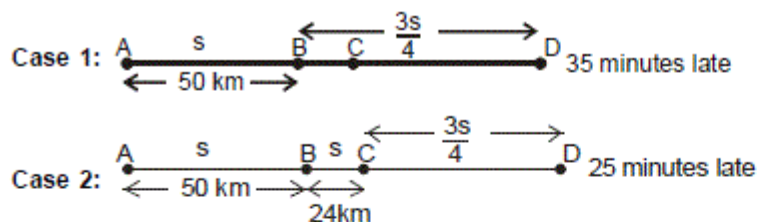
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**Q.58 [11594329]**

A businessman was getting late for a meeting. After he drove 50 km, he closely escaped an accident. So he reduced his speed to  $3/4^{\text{th}}$  of his original speed due to which he got late by 35 minutes for the meeting. Had he escaped that accident 24 km further then he could have saved 10 minutes from 35 minutes. Find the original speed by which the businessman was travelling with?

1 ☐ 48 kmph2 ☐ 54 kmph3 ☐ 36 kmph4 ☐ 58 kmph

**Solution:****Correct Answer : 1**[Answer key/Solution](#)

In case 1 and case 2, the only difference is in stretch BC.

In case 1, the distance BC is travelled with speed  $\frac{3s}{4}$  whereas in case 2, it has been travelled with speed  $s$ . That's why we are getting the time difference of 10 minutes in the two cases.

Now, the ratio of speeds for distance BC in case 1 and case 2 =  $\frac{3s}{4} : s = 3 : 4$

$\therefore$  Ratio of time to travel BC in case 1 and case 2 =  $4 : 3$

As the time difference between these two is 10 minutes.

$\therefore$  The time taken to cover the distance BC in case 1 and case 2 are 40 minutes and 30 minutes respectively.

In case 2, BC is covered with usual speed in 30 minutes or half an hour.

$\therefore$  His original speed =  $\frac{24}{\frac{1}{2}} = 48 \text{ km/hr.}$

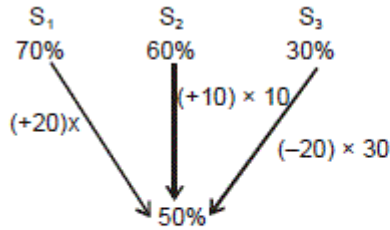
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**Q.59 [11594329]**

There are 3 solutions of alcohol and water, with alcohol content being 70%, 60% and 30%. If 10 litres of solution with 60% alcohol, 30 litres of solution with 30% alcohol and some amount of solution with 70% alcohol, are mixed to obtain a solution having at least 50% alcohol content, which of the following could never be the amount of solution with 70% alcohol in it?

1 ☐ 25 litres2 ☐ 30 litres3 ☐ 35 litres4 ☐ 20 litres

**Solution:****Correct Answer : 4**[Answer key/Solution](#)

For resultant to 50% alcohol content,  
 $20x + (10)10 + (-20) \times 30$  should be zero  
 $\Rightarrow x = 25$

For resultant to be more than or equal to 50% alcohol content,  $x \geq 25$ .

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FeedBack

**Q.60 [11594329]**

The roots of the quadratic equation  $x^2 - 12x + p = 0$  are rational. If  $p$  is a non-negative integer, then how many values can  $p$  take?

**Solution:****Correct Answer : 7**[Answer key/Solution](#)

$$\begin{aligned}
 x^2 - 12x + p &= 0 \\
 \text{For roots to be rational, discriminant (D) should be a perfect square.} \\
 D &= b^2 - 4ac \\
 &= (12)^2 - 4(1)(p) \\
 &= 4(36 - p)
 \end{aligned}$$

should be a perfect square

Non-negative integer values of  $(p)$  are:  
 0, 11, 20, 27, 32, 35 and 36;  
 Hence, number of values of  $p = 7$ .

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FeedBack

**Q.61 [11594329]**

When the length of each side of a triangle is subtracted from the sum of the lengths of the other two sides then, 12 cm, 16 cm and 36 cm, respectively are obtained. Find the area (in  $\text{cm}^2$ ) of the triangle.

1 ☐  $96\sqrt{3}$ 2 ☐ 2563 ☐  $144\sqrt{2}$ 4 ☐ 216



**Solution:****Correct Answer : 1**[Answer key/Solution](#)

Let a, b and c be the sides of the triangle.

$$a + b - c = 12 \dots (i)$$

$$a + c - b = 16 \dots (ii)$$

$$b + c - a = 36 \dots (iii)$$

$$\text{The area of the triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{where, } s = \frac{a+b+c}{2}$$

$$\text{Area} = \sqrt{\left(\frac{a+b+c}{2}\right)\left(\frac{b+c-a}{2}\right)\left(\frac{a+c-b}{2}\right)\left(\frac{a+b-c}{2}\right)}$$

Adding (i), (ii) and (iii)

$$a + b + c = 64$$

$$\text{Hence, the area of the triangle} = \sqrt{\left(\frac{64}{2}\right)\left(\frac{36}{2}\right)\left(\frac{16}{2}\right)\left(\frac{12}{2}\right)} = 96\sqrt{3} \text{ cm}^2.$$

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FeedBack

**Q.62 [11594329]**

Among 130 tourists, 50% of the tourists travel by car and 60% of the tourists travel by bus. Also, exactly 40% of tourists who travelled by only bus visit spot A on first day of their tour. What could be possible number of tourists who visited spot A on the first day of the tour?

1 ☐ 282 ☐ 253 ☐ 154 ☐ 6
☒



**Solution:****Correct Answer : 39** Answer key/Solution

Let usual time = T

Since work = constant  $\Rightarrow$  Time  $\propto \frac{1}{\text{Efficiency}}$ If efficiency becomes  $\frac{13}{10}$ th  $\Rightarrow$  Time becomes  $\frac{10}{13}$ th.

$$\Rightarrow T - \frac{10}{13}T = 30 \text{ min} \Rightarrow T = 130 \text{ min.}$$

If efficiency is  $\frac{10}{13}$ th  $\Rightarrow$  Time =  $\frac{13}{10}$ th of the usual

$$\Rightarrow \frac{13T}{10} - T = \frac{3T}{10} = \frac{3}{10} \times 130 = 39 \text{ min.}$$

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FeedBack

**Q.65 [11594329]**

149 consecutive multiples of 4 are written in ascending order. Find the average of these in terms of 'A', if it is known that the average of the last 53 terms is 'A'.

1 ☐ A - 1922 ☐ A - 2403 ☐ 2A - 524 ☐ A / 64**Solution:****Correct Answer : 1** Answer key/Solution $t_1, t_2, t_3, \dots, t_{149}$ 

As all are consecutive multiple of 4 in ascending order the sequence is an A.P.

Thus, the middle term will be the average i.e.,  $t_{75}$ .

Now, the average of last 53 terms is 'A'.

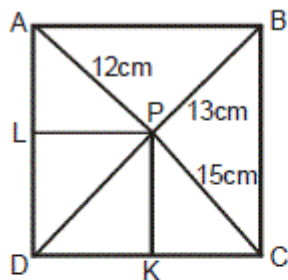
 $t_{97}, t_{98}, t_{99}, \dots, t_{149}$ These are also in AP, so the average will be the middle term i.e.,  $t_{123} = A$ .Now, term  $t_{122} = A - 4$ ,  $t_{121} = A - 8$ , .... as this AP has common difference of 4.So, term  $t_{75} = A - 4 \times (48) = A - 192$ .

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FeedBack

**Q.66 [11594329]**

Let P be a point inside a square ABCD, such that PA = 12 cm, PB = 13 cm and PC = 15 cm. If a square is drawn with PD as diagonal, then what is the length (in cm) of the side of this square?

1 ☐ 52 ☐ 123 ☐ 104 ☐  $5\sqrt{2}$ **Solution:****Correct Answer : 3**[Answer key/Solution](#)

$$\begin{aligned}
 PA^2 + PC^2 &= PB^2 + PD^2 \\
 \Rightarrow 12^2 + 15^2 &= 13^2 + PD^2 \\
 \Rightarrow 144 + 225 &= 169 + PD^2 \\
 \Rightarrow PD^2 &= 200 \\
 \Rightarrow PD &= 10\sqrt{2} \text{ cm}
 \end{aligned}$$

$$\text{Hence, side of the square PKDL} = \frac{PD}{\sqrt{2}} = \frac{10\sqrt{2}}{\sqrt{2}} = 10 \text{ cm.}$$

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FeedBack

**Q.67 [11594329]**

Three natural numbers  $p$ ,  $q$ , and  $r$  are in AP where  $p < q < r$ , whereas  $6p$ ,  $15q$ ,  $40r$  are in geometric progression. Which of the following could be the sum of  $p$ ,  $q$  and  $r$ ?

1 ☐ 542 ☐ 603 ☐ 754 ☐ 80

**Solution:****Correct Answer : 2** Answer key/Solution

Since  $6p, 15q, 40r$  are in GP.

Therefore,  $240pr = 225q^2$  or  $16pr = 15q^2$

Since  $p, q$ , and  $r$  are in AP,  $p = q - d$ ,  $r = q + d$  where  $d$  is the common difference.

So  $16(q^2 - d^2) = 15q^2$

or  $q = 4d$ , and  $p = 3d$ ,  $r = 5d$

Thus,  $p + q + r = 12d$  (or a positive multiple of 12).

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FeedBack

**Q.68 [11594329]**

Once a Goldsmith divided a gold coin into four pieces with weights in the ratio of  $1 : 2 : 3 : 4$ . When the pieces were sold, he got Rs.1,40,000 less. If the cost of gold varies directly as the square of its weight, then find the original price (in Rs. lakh) of the gold coin.

**Solution:****Correct Answer : 2****Your Answer : 200000** Answer key/Solution

Let the original weight of the gold coin be  $10x$ .

Hence, its original price will be  $k(100x^2)$ , where  $k$  is a constant.

The weights of the gold pieces after division are  $x, 2x, 3x$  and  $4x$ . Therefore, their prices will be  $kx^2, 4kx^2, 9kx^2$  and  $16kx^2$ , respectively.

So, the total price of the pieces =  $(1 + 4 + 9 + 16)kx^2 = 30kx^2$ .

Hence, the difference in the price of the original gold and its pieces =  $100kx^2 - 30kx^2 = 70kx^2 = 140000$ .

Hence,  $kx^2 = 2000$  and the original price =  $100kx^2 = 100 \times 2000 = 200000 = \text{Rs. 2 lakh}$ .

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**Q.69 [11594329]**

Let  $f(x) = ax^2 + bx + c$ , where  $a, b$  and  $c$  are certain constants. It is known that  $f(-6) = -6f(8)$ , and 9 is a root of  $f(x)$ . If  $f(13)$  is negative, then  $f(3)$  must be

1 ☐ Greater than zero2 ☐ Less than zero3 ☐ Exactly zero or greater than zero4 ☐ Cannot be determined

**Solution:****Correct Answer : 1**[Answer key/Solution](#)

A quadratic function can be written in terms of its roots as:

$$f(x) = a(x - \alpha)(x - \beta) \quad \text{where } \alpha \text{ and } \beta \text{ are the roots}$$

Since 9 is a root  $f(x) = a(x - 9)(x - \beta)$

It is known that  $f(-6) = -6f(8)$

Or  $a(-6 - 9)(-6 - \beta) = -6a(8 - 9)(8 - \beta)$

Solving we will get  $\beta = -2$

Thus  $f(x) = a(x - 9)(x + 2)$

$f(13) = a(13 - 9)(13 + 2) = 60a < 0$

Thus  $a < 0$

$f(3) = a(3 - 9)(3 + 2) = -30a > 0$  (as  $a < 0$ ).

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FeedBack

**Q.70 [11594329]**

If  $\frac{p}{q+r+s} = \frac{q}{p+r+s} = \frac{r}{q+p+s} = \frac{s}{q+r+p} = k$ , then what could be the possible value of  $\log(3k)$ ?

1 ☐ 1/32 ☐ 13 ☐ 04 ☐ None of these**Solution:****Correct Answer : 3**[Answer key/Solution](#)**Your Answer : 3**

$$\frac{p}{q+r+s} = \frac{q}{p+r+s} = \frac{r}{q+p+s} = \frac{s}{q+r+p} = k$$

$$p = k(q + r + s)$$

$$q = k(p + r + s)$$

$$r = k(q + p + s)$$

$$s = k(q + r + p)$$

$$p + q + r + s = 3k(p + q + r + s)$$

Therefore,  $k = 1/3$  is possible value.

Hence,  $\log(3k) = 0$ .

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